IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 5 June 2025, accepted 21 July 2025, date of publication 29 July 2025, date of current version 4 August 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3593523

RESEARCH ARTICLE

Optimization and Management of Data Center Networks: A Scoping Review on Key Themes, Challenges, and Artificial Intelligence and Machine Learning Approaches

KOGUL SRIKANDABALA[1], (Graduate Student Member, IEEE), KIRISHNNI PRABAGAR[1], (Graduate Student Member, IEEE), SHALINKA JAYATILLEKE[1], PEARLIE ZHANG[2], RICHARD ELLERBROCK[2], SEAN RINAS[2], DASWIN DE SILVA[1], (Senior Member, IEEE), AND DAMMINDA ALAHAKOON[1], (Member, IEEE)

[1]Centre for Data Analytics and Cognition, La Trobe University, Melbourne, VIC 3086, Australia
[2]NEXTDC Ltd., Macquarie Park, NSW 2113, Australia

Corresponding author: Kogul Srikandabala (s.kogul@ltu.edu.au)

ABSTRACT Data Center Networks (DCNs) play a critical role in enabling scalable, efficient, and reliable digital infrastructure. This scoping review presents recent advancements in optimization and management strategies for DCNs, with a particular focus on the application of Artificial Intelligence (AI) and Machine Learning (ML) techniques. This study follows a structured methodology aligned with the PRISMA-ScR protocol and extends it using Large Language Models (LLMs) to support thematic classification and relevance screening. The analysis identified seven principal research domains: optical networking, congestion control, flow scheduling, load balancing, Software-Defined Networking (SDN), fault tolerance, and energy-efficient resource management. Within these domains, this study contrasts traditional approaches with emerging AI-driven methods, highlighting their potential and limitations. Notably, the findings revealed a lack of integrated end-to-end AI/ML frameworks capable of addressing the multifaceted demands of modern DCN environments. Additionally, this review highlights the need for scalable, explainable, and context-aware solutions that align with evolving DCN requirements. By analyzing the current body of knowledge and highlighting critical research gaps, this study provides a comprehensive foundation for future investigations into intelligent DCN design and operation.

INDEX TERMS Artificial intelligence, data center networks, machine learning, network management, optimization, software-defined networking.

I. INTRODUCTION

In the digital age, there is a surge in the demand for network traffic through streaming services that provide on-demand access to entertainment, e-commerce platforms that enable millions of real-time transactions, and social media networks that link billions of people globally with minimal delays in communication. The smooth operation of these services depends on a complex network of interconnected computers and hardware maintained in a large-scale infrastructure. These facilities provide data availability, rapid processing, and effective communication, thereby forming the backbone of the Internet. Maintaining the speed, reliability, and scalability of these infrastructures is becoming increasingly important as society becomes more dependent on digital technology. From guaranteeing continuous connectivity to

The associate editor coordinating the review of this manuscript and approving it for publication was R. K. Tripathy.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

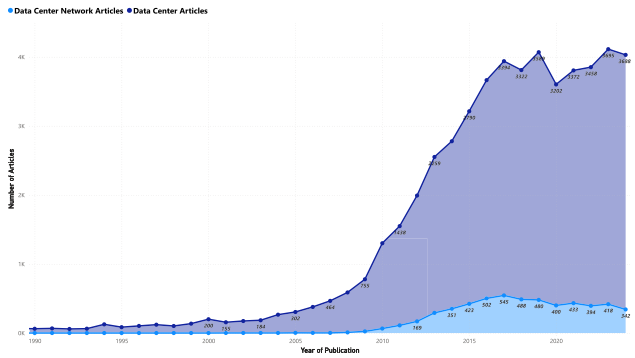


FIGURE 1. Number of articles published on *DCs* and *DCNs* published during the period of 1990-2024. Research in both these areas appeared to have gradually increased over the years but the research within *DCNs* has not evolved at the same pace, thus highlighting a significant gap in this field of research.

allowing fast data processing for AI, these services are critical for modern technological advancements. These facilities, known as Data Centers (DCs), form the foundation of modern computing and communication. DCs accommodate several computing resources that collect, store, share, manage, and distribute large volumes of data. They consist of all the necessary DC facility elements (space, power, and cooling) and IT infrastructure elements (servers, storage, and networks), based on business requirements [1].

DCs comprise various interconnected components, among which the DCN functions as a central brain. DCNs are specifically engineered networks that interconnect many components within a DC and play a crucial role in day-to-day operations [2]. Servers and storage systems within a DC can realize their full potential only if they can exchange data at extremely high speeds, maintain low latency, and experience minimal downtime. Consequently, the underlying network is vital for meeting the operational requirements of cloud platforms, streaming services, and enterprise applications. However, when contrasting DCNs with conventional computer networks, such as Local Area Networks (LAN), DCNs present a distinctive set of challenges that must be addressed. Xia et al. highlighted key challenges in DCNs, including their massive scale, diverse range of applications, high energy consumption, and the need to maintain rigorous Service Level Agreements (SLAs) [2]. Previous studies have attempted to address these challenges, as illustrated by the increasing research trends in Figure 1. However, in the past decade, technological progress in server hardware has been driven by Moore's law [3], leading to exponential increases in processing power, memory capacity, and energy efficiency. Although computational capabilities have advanced rapidly, DCNs have not evolved at the same pace. Figure 1 illustrates the research gap observed over the years. The increasing density and speed of servers have placed greater demands on network infrastructure; however, traditional networking architectures often struggle to keep up, leading to bottlenecks, latency issues, and inefficient data transmission. These challenges highlight the growing complexity of DCNs and emphasize the need for continuous research and innovation to address these issues. Although advancements in server hardware have been rapid, DCN architectures and management strategies have not kept pace, leading to significant bottlenecks. Understanding how research has evolved to address these gaps is critical for identifying key areas for further exploration.

A. RELATED SURVEYS AND REVIEWS

A considerable body of research has surveyed and reviewed various aspects of DCNs. These reviews cover a broad range of topics, including congestion-control solutions, traffic management mechanisms, and optimization strategies. With the increasing complexity and scale of modern DCNs, researchers have sought to understand and enhance network performance using innovative methodologies and frameworks. Many surveys have also highlighted the emergence of AI and ML as promising tools for automating and optimizing DCN operations, bringing new dimensions to traditional network management approaches.

Recent surveys on DCNs have explored various critical themes, each addressing specific challenges and advancements in the field. These themes can be categorized as follows:

- **Optimization Strategies in DCNs:** Network optimization remains a focal point of many studies, with researchers exploring methods such as load balancing, traffic engineering, and routing mechanisms.
- **Congestion Control & Traffic Management:** The exponential growth in data traffic within DCNs has made congestion control and traffic management essential topics of study. Several surveys have provided comprehensive discussions on congestion control algorithms, network flow management techniques, and their practical implementation. These surveys compared traditional methods with emerging AI-driven solutions, emphasizing their impact on improving data throughput and reducing packet loss.
- **Data Aggregation & Processing:** The demand for real-time data analytics in large-scale DCs has led to a growing interest in data aggregation techniques. Some surveys have investigated in-network processing and data aggregation methodologies that enhance scalability and computational efficiency. These studies focused on distributed computing frameworks and their ability to streamline data flows within high-performance networks.
- **Converged & High-Performance Networks:** Advancements in hardware and software technologies have resulted in the development of converged network infrastructures that integrate multiple networking paradigms. Surveys on high-performance networking discuss innovations, such as fiber-optic interconnects, software-defined networking (SDN), and network function virtualization (NFV). These studies examined

TABLE 1. Survey and Review Papers on DCNs (2010-2024): This table categorizes survey papers on DCNs from 2010 to 2024 across five themes: (1) Converged and High-Performance Networks, (2) Optimization Strategies, (3) Congestion Control and Traffic Management, (4) Data Aggregation and Processing, and (5) Next-Generation Network Technologies. Numbers indicate relevant surveys, while empty cells indicate gaps in the literature, highlighting research trends over time.

Year	Converged and High Performance Networks	Optimization Strategies in DCN	Congestion Control and Traffic Management	Data Aggregation and Processing	Next Generation Network Technologies
2010	[4]–[6]	-	-	[7]	-
2011	-	-	-	-	-
2012	[8,9]	-	[10]	[11,12]	-
2013	[13,14]	-	[15]	[16,17]	[18]–[21]
2014	[22,23]	-	[24,25]	[26]–[29]	[30]–[32]
2015	[33]	[34]	-	[35]–[40]	[41]
2016	[42]	-	[43]	[44]–[46]	[47,48]
2017	-	-	[49]	[50]–[52]	[53]–[55]
2018	[56]–[60]	[61]–[63]	-	[64]	[65,66]
2019	[67]–[69]	-	[70]	[71]–[74]	[75]–[78]
2020	[79]	-	[80]	[81,82]	[83]
2021	[84,85]	[86]	[87,88]	[89,90]	[91]–[95]
2022	[96]	[97]	-	[98,99]	-
2023	[100]–[102]	-	[103,104]	[105]	-
2024	[106]	[107]	-	-	-
Total	**27**	**7**	**12**	**33**	**25**

the benefits and challenges of transitioning to fully converged network architectures and their implications for large-scale DCs.

- **Next Generation Network Technologies:** The rapid evolution of DCNs has driven research into emerging technologies that enhance scalability, efficiency, and security. Surveys in this domain explore wireless DCNs, such as 60 GHz networks, for high-speed, low-latency communication, and energy-efficient networking strategies to optimize power consumption. Other studies have focused on software-defined networking (SDN) and hyperscale architectures that leverage optical switching and automation to improve agility and performance. These advancements address the key challenges in cloud and edge computing, ensuring that DCNs remain adaptable and future-ready in the face of growing data demands.

A more detailed breakdown of the survey papers covering these themes from 2010 to 2024 is provided in Table 1. This table categorizes past research efforts and highlights key trends in DCN-related studies. As shown in Table 1, while certain areas, such as Data Aggregation and Processing, have received substantial attention, other aspects, including Optimization Strategies in DCNs, remain comparatively underexplored.

Existing surveys have provided valuable insights into various aspects of DCN optimization, including congestion control, traffic management, and next-generation network technologies. However, these studies often focused on the isolated aspects of DCNs rather than offering a comprehensive synthesis. Furthermore, despite the increasing role of AI/ML in network optimization, few studies have systematically examined their application in DCN management. This creates an opportunity to conduct a more structured and holistic review that addresses gaps in the literature.

Several surveys and reviews have examined various aspects of DCNs. Some studies have explored the potential of AI and ML in networking, particularly for automating and optimizing DCNs. However, we have identified the following gaps within these surveys and reviews,

- *Narrowed focus on specific aspects of the DCN* - Most studies only cover one aspect of DCNs, such as network topology design, congestion control, or energy efficiency. These studies provide significant insights but do not capture the full scope of DCN research. Thus, a scoping review capturing research in all areas of DCN would be helpful.
- *Lack of systematic and scoping review approaches* - Among the few reviews that discuss all the applications within DCNs, the majority adopt narrative review methodologies rather than a systematic or scoping review framework. Consequently, their coverage is often narrow, focusing on specific techniques or isolated case studies rather than providing a structured synthesis of the entire research landscape. A systematic scoping review is needed to map, categorize, and analyze the existing body of literature, ensuring that the breadth of research is adequately captured and that gaps in DCN studies are clearly identified.
- *Limited analysis of AI and ML in DCN optimization and management* - Although research on DCN architectures and optimization is extensive, most reviews overlook AI/ML applications in network efficiency, traffic management, and automation. As AI-driven solutions become integral to network operations, this gap limits our understanding of how intelligent automation and

predictive analytics can enhance performance, reduce energy consumption, and improve reliability.

B. CONTRIBUTIONS

To bridge the gaps identified in Section I-A, this study presents a scoping review of DCN research, providing a structured analysis of key themes, methodologies, and emerging trends. By incorporating AI/ML-focused discussions and employing a systematic scoping approach, this review aims to offer a comprehensive understanding of how DCNs can be optimized and managed efficiently.

The main contributions of this study are as follows.

1) A scoping review conducted following the PRISMA-ScR guidelines [108], ensuring a systematic and transparent approach to reviewing DCN research.
2) A novel LLM-based approach for screening and analyzing studies, enabling efficient topic modeling and literature review in large-scale research where manual examination is impractical.
3) Identification and categorization of key research themes in DCNs, providing insights into the primary focus areas and trends within the field
4) A detailed analysis of major challenges in DCN research and an evaluation of proposed solutions, offering a structured overview of how researchers are addressing critical issues.
5) An exploration of fundamental network concepts utilized in DCN research, highlighting their role in shaping network optimization and solution development.
6) An investigation of AI/ML techniques in DCN solutions, including an assessment of implementation challenges and a comparative evaluation against traditional approaches.

By mapping the landscape of DCN research and identifying critical gaps, this study lays the foundation for future research on AI-driven network optimization. The remainder of this paper is organized as follows: Section II presents the background and motivation for this review, establishing the context of DCNs and the importance of AI-driven optimization. Section III outlines the research questions and objectives that guided this scoping review. Section IV describes the methodology, including the literature search strategy, selection criteria, and data extraction process. Section V provides an overview of the foundational concepts of DCNs, including network architectures, protocols, and key performance metrics. Section VI reviews related survey papers and highlights their limitations, thereby positioning the unique contributions of this study. Section VII introduces a novel taxonomy for categorizing research on AI-driven DCN optimization, organizing studies by focus area, and methodological approach. Section VIII discusses the main thematic areas identified in the literature. Finally, Sections IX–XI present a comprehensive discussion, synthesizing the main insights, outlining the current challenges, and proposing future research directions.

II. METHODS

This section outlines the methodology employed in this review. This study followed the PRISMA-ScR framework, incorporating both manual and automated techniques to ensure systematic and unbiased selection of studies. The review process included the formulation of research questions, identification and screening of relevant literature, application of inclusion and exclusion criteria, and data extraction for analysis. Given the large volume of retrieved studies, automation techniques, including Large Language Models (LLMs), were leveraged to enhance efficiency, particularly in thematic screening. The following subsections provide a detailed breakdown of this methodology.

A. RESEARCH DESIGN

This scoping review is designed to answer the following research questions:

- **RQ1** - What are the key research themes in DCN and how do they contribute to the optimization and management of DCN?
- **RQ2** - What are the main challenges within each of these themes and what solutions/breakthroughs have been proposed to address them?
- **RQ3** - What fundamental network concepts support the implementation of these solutions, and how do they enable DCN optimization and management?
- **RQ4** - How do AI/ML techniques leverage network concepts to address DCN challenges, and how do they compare to traditional approaches?
- **RQ5** - What are the main challenges in designing, developing and deploying AI/ML-based solutions for the optimization and management of DCN?

The answers to these questions are discussed in detail in the following section.

B. IDENTIFICATION AND SCREENING OF RELEVANT STUDIES

The identification and screening processes followed the PRISMA-ScR process, and the methodology is illustrated in Figure 2. This approach is widely used in medicine and biology. The review methodology was adapted for application in the context of this study. This methodology has three unique phases that occur serially, as illustrated in the diagram.

A comprehensive search was conducted across multiple academic databases using the keywords *"Data Center Network"* and *"Data Centre Network"* to ensure the inclusivity of both American and British English terminology. The search retrieved a large number of studies that were processed for duplicate removal to eliminate redundancy. In addition, inaccessible articles were excluded to ensure that only retrievable and relevant studies were retained for further screening. This step was crucial for establishing a robust dataset while maintaining consistency in the study selection.

Given the large number of retrieved studies, manual screening of all records was not feasible. While certain

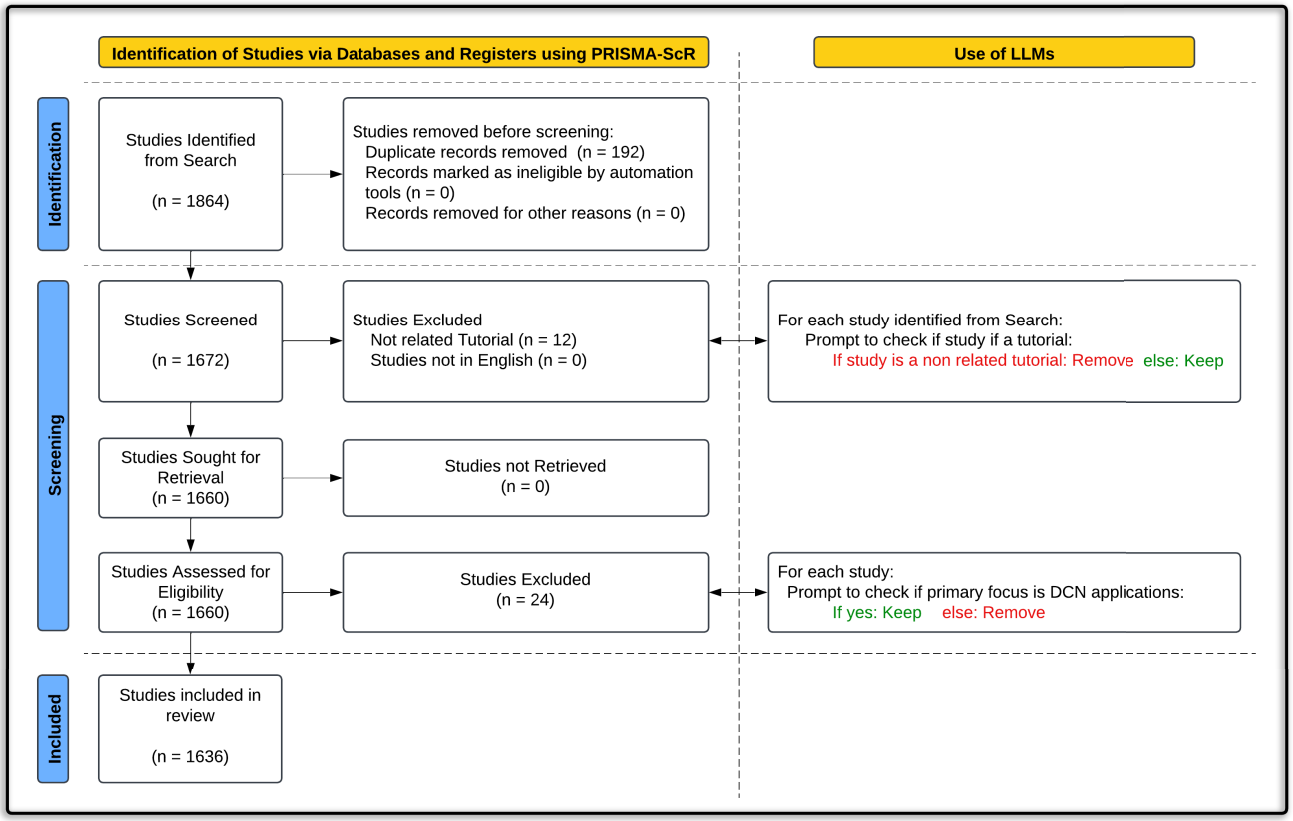


FIGURE 2. **The PRISMA-ScR methodology followed during the study. The methodology was further extended using LLMs because the number of studies screened was too large for manual filtering. The methodology consisted of three stages: 1) Identification: The initial search returned 1,864 records, of which 192 were removed for duplication and other specified reasons before progressing to the screening stage. 2) Screening: A total of 1,672 studies were screened, with 12 unrelated tutorials excluded while ensuring that all were in English. After this step, 1,660 studies were retrieved, and all were successfully accessed. 3) Eligibility and Inclusion: After assessing 1,660 studies for eligibility, 24 were excluded, leaving 1,636 studies included in the final review. LLMs were used in both screening and eligibility stages to automate filtering, reducing manual workload while ensuring relevance to DCN applications.**

objective criteria, such as access and publication types, can be reliably assessed using metadata, other aspects, particularly thematic relevance, require deeper evaluation. To ensure efficiency without compromising accuracy, a hybrid approach combining automation through Large Language Models (LLMs) and manual validation was adopted.

The large number of studies in the screening phase made manual analysis unfeasible, necessitating automation for the efficient identification of relevant studies. Our goal was to eliminate bias while achieving human-level accuracy. Large Language Models (LLMs) have been widely used in literature reviews across disciplines. Dennstädt highlighted their role in automating title and abstract selection in biological research [109]. Mostafapour demonstrated how LLMs, such as ChatGPT, assist researchers by generating early drafts, outlines, and streamlining the review process, especially for those with limited resources [110]. Hasan explored LLM integration into systematic reviews, particularly for risk of bias assessment, to enhance review rigor [111]. Chelli further emphasized LLMs' ability of LLMs to improve literature searches and consolidate findings, making them invaluable for automated screening [112]. These studies confirm that LLMs are crucial for automating and improving the literature review process. Given the scale of studies in this scoping review, we will leverage LLMs for thematic identification, reducing manual workload while maintaining accuracy.

Table 2 outlines the selection criteria used in this scoping review. The studies were evaluated on the basis of the following five key criteria: main theme, language, access type, publication type, and study type. The applicability of automation varies across different screening criteria.

- *Access Type* and *Publication Type*: These criteria can be reliably assessed using databases like CrossRef, which provides metadata that can be used to assess whether a study is open-access and peer-reviewed. This step can be fully automated using a Python script.
- *Language* and *Study Type*: While word-matching techniques can identify English-language studies and distinguish between surveys, reviews, and original research, some manual inspection is still required due to potential inconsistencies in metadata and classification.

- *Main Theme*: Identifying whether a study explicitly focuses on DCNs is the most challenging step. Keyword-based filtering alone is insufficient because many studies mention DCs but do not focus on DCNs. This step requires a manual review. However, given the scale of the dataset (thousands of studies), a Large Language Model (LLM)-based approach is necessary to automate thematic analysis while reducing human effort.

By integrating the CrossRef API for metadata-based filtering, word matching for language and study type classification, and LLMs for thematic identification, the screening process balances efficiency and accuracy. These automated methods significantly reduce the manual workload while maintaining high screening precision. With the final dataset curated, the next step involved extracting key insights from the selected studies.

Seven key themes were extracted from the filtered papers, which collectively represented the different focus areas investigated by research published in the domain of DCNs.

III. MAJOR RESEARCH THEMES

The major themes identified in the literature are as follows:

- **Optical Networking and High-Speed Transmission (N=328, 20%):** With the exponential growth of data traffic demand, Optical Networking has become the solution offering ultra-fast transmission rates, reduced latency, and lower power consumption.
- **Congestion Control and Latency Optimization (N=233, 14%):** With high-speed transmission, the challenge of managing network congestion and reducing latency arises. Congestion control mechanisms are essential for preventing packet loss, minimizing delays, and optimizing traffic flow across the entire network.
- **Flow Scheduling and Deadline-Aware Networking (N=216, 13%):** Not all network traffic is equal, some applications require low latency and guaranteed deadlines. Flow scheduling is used to ensure the efficient allocation of network resources by prioritizing critical workloads and application-specific deadlines.
- **Load Balancing and Traffic Distribution (N=123, 8%):** Effective flow scheduling creates the need for load balancing to optimize resource utilization in order to evenly distribute network traffic,yu thereby preventing resource over-utilization and under-utilization.
- **Software-Defined Networking (SDN) and Traffic Control (N=198, 12%):** The evolution of load balancing techniques necessitates programmable, flexible traffic control mechanisms. SDN enables centralized network management, dynamic traffic engineering, and real-time policy adjustments.
- **Fault Tolerance and Network Resilience (N=250, 15%):** The implementation of SDN brings new challenges in ensuring network reliability. Fault-tolerance mechanisms enhance network resilience by implementing redundancy strategies, failure recovery, and predictive maintenance techniques.
- **Energy-Efficient and Cloud Resource Management (N=288, 18%):** Optimizing network performance is incomplete without addressing sustainability. Energy-efficient strategies ensures sustainability by optimizing power consumption in large-scale cloud environments

The following sections synthesize the filtered studies, categorized by their themes, to extract meaningful insights into the challenges that drive the research, contributions made by the research, emerging trends, and the effectiveness of various solutions proposed. Figure 3 presents a quantitative analysis of the distribution of themes across these factors.

Discussion on RQ1

What are the key research themes in DCN and how do they contribute to the optimization and management of DCN?

The main themes in DCN research are the seven ones that are highlighted above. Together, they provide a basis for current DCN studies and innovations. By addressing important issues in terms of speed, efficiency, adaptability, and reliability, these main themes can help optimize and manage DCNs. Whereas congestion control and latency optimization guarantee an uninterrupted flow of network traffic free from bottlenecks, optical networking guarantees a high-speed data transfer in DCNs. Critical workloads are prioritized in flow scheduling and deadline-aware networking, thereby enhancing service quality. Load balancing and traffic distribution optimize resource usage to prevent system overload. While fault tolerance and network resilience improve dependability by reducing service disruptions, software-defined networking (SDN) and traffic management add programmability and flexibility, allowing real-time traffic adjustments. Finally, energy-efficient cloud resource management guarantees sustainability by lowering operating expenses and power consumption. Taken together, these components determine the strategic direction in which DCNs will evolve.

IV. DRIVING FORCES BEHIND DCN OPTIMIZATION: CHALLENGES ACROSS KEY THEMES

Building upon the thematic overview presented in Section III, this section delves deeper into each of the seven research themes by systematically examining the key challenges, underlying motivations, and unresolved issues that drive ongoing optimization efforts in DCN research.

The evolution of DCNs is driven by challenges such as scalability, congestion, security risks, and energy consumption. These challenges act as driving factors that shape the purpose of the DCN research. This section explores the key driving forces behind DCN optimization and examines its purpose, motivations, and challenges across the seven fundamental themes extracted from the literature.

Optical Networking and high-speed transmission are critical for ensuring seamless data flow in DCNs. Traditional architectures struggle to keep up with rising data traffic, creating the need for intelligent traffic engineering, energy-efficient optical components, and resilient security mechanisms. Research has focused on AI-driven traffic engineering,

TABLE 2. This table outlines the selection criteria applied in this scoping review of DCNs. Studies were evaluated based on relevance, language, access type, publication type, study type, and publication year.

Criteria	Inclusion	Exclusion	Reason
Main Theme	Studies focused on DCNs and their applications.	Studies related to DCs, but not specifically focusing on DCNs.	To ensure that the review specifically addresses the core topic, i.e., DCNs, and avoids unrelated content.
Language	Studies published in English.	Studies published in languages other than English.	Non-English studies could not be screened by the authors due to language limitations.
Access	Open-access studies.	non-open-access studies.	To allow easy access to the full text of the selected studies.
Publication Type	Peer-reviewed journal articles and conference papers.	Non-peer-reviewed studies (e.g., white papers, preprints, blog posts, and industry reports).	To ensure the credibility and scientific rigor of the included studies.
Study Type	Review papers, survey papers, and original method papers.	Other types of papers that do not fall into these categories.	To cover a broad but relevant range of research on DCNs.

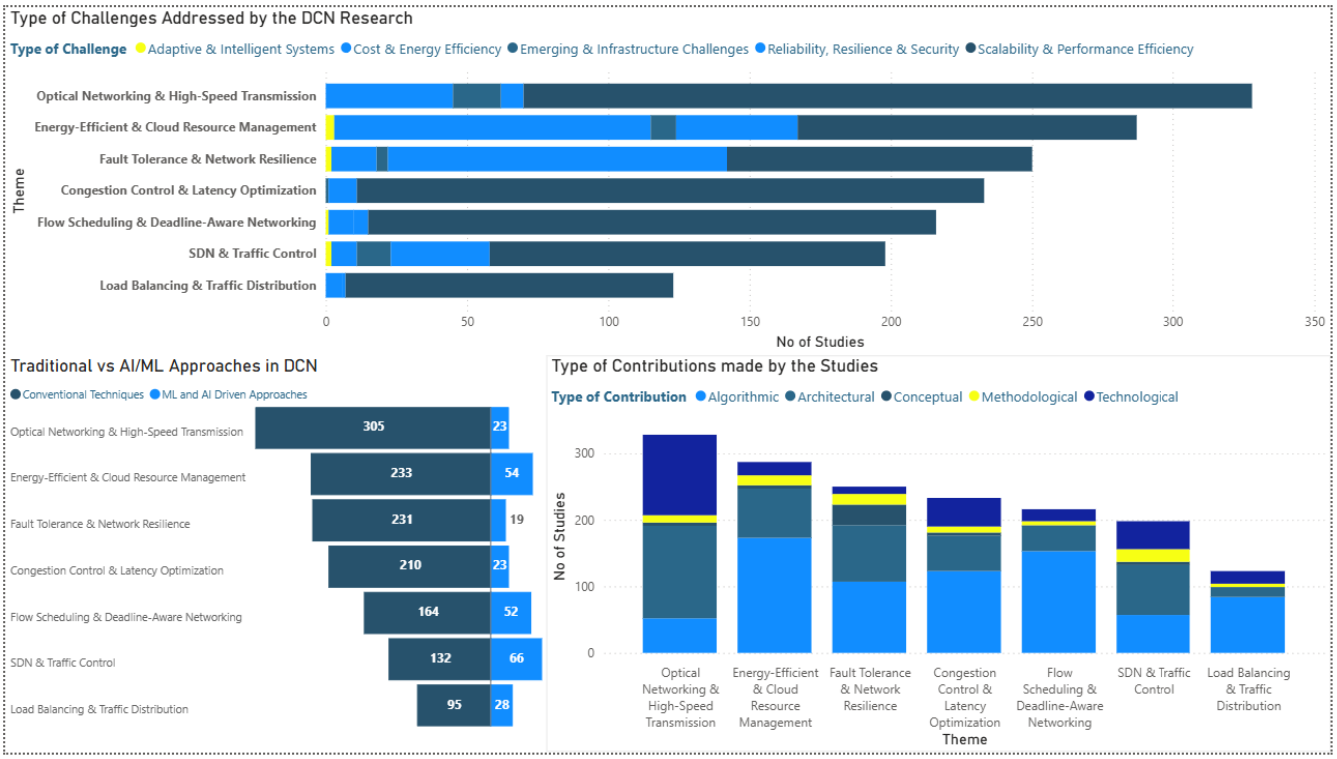


FIGURE 3. Dashboard presents the distribution of research themes categorized by (1) Type of challenges addressed, (2) AI vs. traditional approach in DCN research, and (3) Type of contributions.

multilayer traffic optimization, and ML-based predictive congestion control to enhance network performance [113], [114]. Advancements in energy-efficient optical components aim to reduce operational costs, whereas encryption and fault-tolerance mechanisms strengthen network security and resilience [115].

Congestion Control and Latency Optimization are vital for reliable transmission in DCNs. However, static congestion control mechanisms struggle to adapt to dynamic traffic conditions, leading to performance bottlenecks and latency. These challenges create the need for adaptive and proactive transport optimization to enhance network responsiveness. Research has focused on addressing these issues through adaptive congestion control techniques [116] which dynamically adjust the traffic flow to prevent congestion. Latency reduction can be handled by employing predictive and proactive transport strategies [117] and credit-based and RTT-driven approaches [118].

Flow Scheduling and Deadline-Aware Networking are essential for managing increasing traffic loads in DCNs. As traffic loads continue to increase, ensuring low-latency and deadline-aware transmissions is vital for supporting real-time, high-priority traffic. These challenges create the need for efficient traffic scheduling and accurate flow

classification to optimize network performance. To meet these needs, research has explored priority-based scheduling, adaptive flow classification, and dynamic resource allocation as key solutions [119], [120], [121] for improving flow efficiency and ensuring that critical tasks meet their deadlines.

Load Balancing and Traffic Distribution are essential to manage network traffic growth, preventing bottlenecks and load imbalances creating the need for efficient routing strategies. OpenFlow-based routing strategies have been investigated to detect and mitigate elephant flows and improve traffic scalability and distribution [122]. Switch-migration techniques in SDN controllers have also been explored to dynamically reduce congestion and enhance load-balancing [123]. Studies on dynamic distributed multipath load balancing have introduced strategies to ensure that network operations remain uninterrupted during failures [124]. Furthermore, research has focused on creating adaptive load-balancing mechanisms to minimize packet reordering and optimize the QOS [125].

Software-Defined Networking (SDN) and Traffic Control have emerged as a foundational approach to managing the growing complexity of Data Center Networks (DCNs). As traffic volume surges and application demands become increasingly dynamic, traditional network architectures face limitations in terms of scalability, programmability, and adaptability. SDN addresses these issues by decoupling the control and data planes, thereby enabling centralized and programmable control over network behavior. However, this architectural shift introduces new challenges, including optimal controller placement, control-plane scalability, and integration of hybrid SDN models. These issues have been the focus of research exploring telemetry-based monitoring frameworks, intrusion-aware datasets, and RPC-optimized architectures to support efficient and resilient SDN deployment [126], [127], [128]. Furthermore, traffic control within SDN must address emerging security threats. Studies have shown the effectiveness of supervised machine learning models, such as SVM, K-Nearest Neighbors, and Decision Trees, for accurate real-time traffic classification and DDoS attack detection in emulated SDN environments [129], [130]. Further advancements in AI-driven and Moving Target Defense (MTD)-enabled architectures offer additional protection for vulnerable control and data planes [131]. Collectively, these innovations reinforce SDN's role of SDN in shaping scalable, intelligent, and secure DCNs capable of supporting high-throughput, low-latency demands in evolving data-center environments.

Fault Tolerance and Network Resilience are critical for ensuring the reliability and stability of DCNs. However, network breakdowns, undetected failures, and reliability issues continue to pose risks to networks. The occurrence of unpredictable failures in edge computing infrastructure necessitates advanced failure prediction and mitigation strategies to minimize service disruptions [132]. Cost and energy efficiency concerns are addressed by optimizing network architectures and reducing the dependence on expensive wired links, thereby enhancing fault resilience and reducing operational costs [133]. These challenges create the need for intelligent fault-tolerant mechanisms, self-healing networks, and energy-efficient failure-recovery strategies.

Energy-Efficient and Cloud Resource Management is pivotal in optimizing DC operations amidst various challenges. DCs consume vast amounts of power, leading to high operational expenses and environmental impacts, creating cost and energy efficiency challenges [134]. Traditional network resource management struggles to dynamically allocate resources to accommodate growing workloads, causing scalability and performance challenges, as [135]. Concerns about reliability, resilience, and security create the need for solutions to mitigate failures, cyber threats, and disasters that could compromise service availability [136]. These challenges require intelligent energy-aware resource allocation, adaptive workload distribution, and secure cloud management strategies to enhance sustainability and performance. [137].

V. KEY CONTRIBUTIONS AND ADVANCEMENTS IN DCNS

The challenges that fueled DCN research have led to significant contributions across the seven themes in DCN. These contributions can be divided into five categories: conceptual frameworks, methodological innovations, architectural advancements, algorithmic improvements, and technological development. This section explores these key contributions, highlighting the pivotal solutions that have transformed DCNs and set the foundation for future research.

A. CONCEPTUAL FRAMEWORKS

Conceptual frameworks (N=52, 3%) have introduced new theories, paradigms, and foundational models that shape future networking research. Collectively, these contributions provide a strong theoretical foundation for improving the efficiency of DCNs.

In optical networking, the identification of critical security threats in wireless optical networks is a pivotal conceptual advancement that assists in highlighting vulnerabilities for secure high-speed data transmission [138]. Congestion control introduces a comprehensive conceptual study on TCP coexistence and its effects on modern DC traffic, provides vital insights into how network protocols interact, and helps shape smarter traffic routing strategies [139]. Flow scheduling and deadline-aware networking have been significantly improved by the introduction of the coflow age (CA) metric concept, which enhances priority scheduling for latency-sensitive applications [140]. In SDN-control, the most significant conceptual innovation is the development of traffic divergence theory, which enhances our understanding of network traffic flow and enables more efficient SDN-based control mechanisms [141]. The theoretical understanding of how extra edge connectivity strengthens network resilience and aids self-healing has improved our knowledge of network durability [142].

Similarly, the introduction of the cost-dominant fairness concept for bandwidth allocation represents a major breakthrough in optimizing resource distribution in cloud infrastructures [143].

B. METHODOLOGICAL INNOVATIONS

Methodological innovations (N=82, 5%) focus on refining research methodologies, validation techniques, security frameworks, and performance-testing strategies.

Key innovations in optical networking include wavelength bank design methodology, which systematically improves spectral efficiency, and DSP-based signal detection methodologies, which provide structured approaches for identifying transmission errors [144]. Similarly, a methodology integrating cooperative MPTCP models and reinforcement learning techniques was introduced to manage network traffic efficiently, prevent congestion, and adjust data flow in real time to reduce delays [145]. In flow scheduling, new store-and-forward scheduling methods have been created to make better use of bandwidth and reduce processing delays in time-sensitive applications [146]. In load balancing and traffic management, the HF^2T methodology has improved the distribution of network traffic, making it more efficient in changing environments [147]. A significant contribution to fault-tolerant mechanisms includes a methodology involving two-stage queuing models that was introduced to predict network delays, making networks more resilient to failures [148]. Another pivotal novelty is the introduction of new mathematical models to help distribute resources more effectively, thereby cutting costs while maintaining the performance of energy-efficient cloud computing [149].

C. ARCHITECTURAL ADVANCEMENTS

Architectural advancements (N=479, 29%) focused on network design, topology, and physical/logical infrastructure. These contributions often involve novel routing structures and system-level changes that alter how networks are physically and logically structured.

Hybrid optical-wireless architectures integrated with software-defined optical networking increase bandwidth adaptability and transmission speeds [150]. Frameworks such as RTT-based congestion control system & high-speed packet scheduling system [151] minimizes congestions and reduces latency. Architectures such as AIFO's single FIFO queue and multi-layer priority-aware scheduling enhance traffic prioritization and deadline-aware transmission [119]. Frameworks like ConWeave introduces a fine-granularity rerouting system that enhances load balancing in RDMA-based networks [152]. SDN architectures use dedicated RPC processors for microservices, which optimize latency and performance in distributed computing environments [153]. Hierarchical fault-tolerant network architectures have been created to enhance failure recovery mechanisms and improve DCN resilience [154]. AI-based workload migration, cost-efficient virtual network embedding, and energy-aware scheduling architectures reduce power consumption and optimize cloud infrastructure utilization [134].

D. ALGORITHMIC IMPROVEMENTS

Algorithmic improvements (N=749, 46%) involve the development of intelligent algorithms and techniques to optimize network functions. These contributions consist of advanced computing techniques that influence network processes and optimize decision-making.

Transfer learning algorithms have been proposed to enhance link failure prediction and network performance in high-speed optical systems [155]. Self-tuning PI congestion control mechanisms dynamically adjust window sizes to adapt to changing network conditions and minimize congestion [116]. Fine-grained scheduling strategies have been created to prioritize latency-sensitive applications, ensuring efficient bandwidth allocation [119]. Highly efficient switch migration algorithms optimize controller workloads, ensuring balanced traffic flow in SDN-based networks [123]. Advanced SDN-based traffic control mechanisms with DRL-based routing enable adaptive traffic control that reduces congestion and improves packet delivery [156]. Algorithms leveraging deep reinforcement learning (DRL) have been used to create self-healing networks, significantly improving failure prediction and recovery [132]. Novel resource allocation strategies (modified BFS algorithm) and adaptive VM placement strategies have been proposed to effectively schedule tasks in cloud computing environments and optimize resource utilization while maintaining QoS standards [134].

E. TECHNOLOGICAL DEVELOPMENTS

Technological innovation (N=274, 17%) involves new hardware, network devices, transmission and communication technologies, and software advancement.

The major contributions of technology in optical networking are dual-frequency lasers to improve optical communication stability. Clock phase caching, enabling sub-nanosecond recovery in optical networks. Additionally, advanced modulation formats increase the efficiency of optical data transmission [157], [158]. The Scalable Reliable Datagram (SRD) protocol, a newly designed communication enhancement, was created to enhance communication performance [159]. SmartFCT technology leverages deep reinforcement learning (DRL) with SDN to enable real-time traffic management [157]. A software-based multipath RDMA solution with a tangible system, Maestro, was proposed to improve the load distribution efficiency [160]. Another significant contribution was the In-band Network Telemetry (INT), an active system for real-time gray failure detection, making it a directly applicable network resilience solution [161]. KeySFC integrates SDN with strict source routing, providing an optimized, scalable, energy-efficient networking solution [162].

Discussion on RQ2

What are the main challenges within each of these themes?

The primary challenges in DCN research can be categorized into key areas:

Scalability, congestion, latency, security, fault tolerance, and energy efficiency challenges owing to rising data traffic and complex architectures. High-speed optical networking, adaptive congestion control, anomaly detection, intelligent traffic management, energy-efficient resource management, workload migration, and green networking solutions are critical to addressing these issues.

What solutions/breakthroughs have been proposed to address them?

To address these, DCN research has introduced architectural improvements (hybrid networks, fault-tolerant topologies), algorithmic optimizations (deep learning for traffic control, failure prediction), technological innovations (in-band network telemetry, high-speed scheduling), and methodological refinements (traffic modeling, congestion strategies). These contributions enhance performance, security, and sustainability, ensuring scalable and resilient DCNs.

VI. CORE NETWORKING PRINCIPLES IN DCN

Advancements in DCN research rely on a strong foundation of network concepts. To enable intelligent and adaptive DCN operations, researchers have focused on three key concepts: Profiling, Anomaly Detection, and Representation.

A. MODELING AND REPRESENTATION

Representation in a DCN involves modeling network structures, traffic flows, and system behaviors for analysis, simulations, and routing optimization. These models form the basis for advanced DCN solutions.

Improving network representations leads to enhanced performance. In [163], a mathematical model for Clos networks was developed to optimize the DC topology. Similarly, [152] focused on simulating Remote Direct Memory Access (RDMA) behavior to analyze its impact on network performance.

Network representation learning has also been explored for various applications, including traffic optimization and anomaly detection. For example, in [124], a leaf-spine topology was used to efficiently structure DCN data, making it suitable for AI-driven network analysis. Another study [125] leveraged representation learning to optimize routing protocols and reduce packet reordering, leading to more efficient data transmission.

B. NETWORK BEHAVIOR PROFILING

Profiling is crucial for understanding traffic patterns, network behaviors, and resource utilization, thereby enabling the implementation of adaptive DCN management strategies.

Profiling techniques have been used to optimize DCN performance. Research in [123] applied profiling techniques to analyze network traffic and improve switch migration strategies in Software-Defined Networking (SDN). Similarly, in [122], profiling methods were used to simulate network behavior, allowing for better traffic distribution and efficiency. Another study, [125], leveraged profiling to assess packet reordering, which is essential for ensuring Quality of Service (QoS) in DCNs. AI-driven profiling techniques are gaining popularity. In [124], ML models were used to classify network flows dynamically, improving resource allocation.

C. DETECTING ANOMALIES AND SECURITY THREATS

Anomaly detection is a critical component of DCN security and performance management for identifying threats, misconfigurations, and failures that affect network stability.

With the increasing complexity of modern DCs, detecting anomalies in normal network behavior can help mitigate security threats, misconfigurations, and failures. In [164], an AI-based anomaly detection model was developed to detect malicious activity in virtualized environments. Similarly, in [165], deep learning models were trained on SDN controller data to enhance security monitoring.

Anomaly detection also plays a crucial role in defending against cyber threats, such as Distributed Denial-of-Service (DDoS) attacks. Research in [166] focused on real-time traffic monitoring to detect early signs of DDoS attacks and prevent large-scale disruptions. Another study [167] combined statistical methods with network telemetry to detect traffic anomalies in cloud-based DCNs, thereby improving network reliability.

D. INTERDEPENDENCE OF REPRESENTATION, PROFILING, AND ANOMALY DETECTION

Representation forms the foundation by providing structured models of network behavior. Profiling builds on this by analyzing traffic patterns and resource usage and extracting meaningful insights for optimization. Anomaly detection relies on both representation and profiling to identify deviations, detect threats, and enhance security. These three concepts create a continuous feedback loop: representation structures the data, profiling interprets it, and anomaly detection refines both by identifying irregularities. Their integration enables intelligent, adaptive, and automated DCN management to improve the efficiency and security.

Discussion on RQ3

What fundamental network concepts support the implementation of these solutions, and how do they enable DCN optimization and management?

The key concepts are Representation, Profiling, and Anomaly Detection.

The representation structures network behaviors for traffic optimization, routing efficiency, and AI-driven analysis. Profiling examines traffic patterns and resource usage and improves performance, load balancing, and adaptive resource allocation. Anomaly Detection identifies security threats, failures, and misconfigurations, aiding in intrusion detection, DDoS mitigation, and real-time monitoring. These concepts are interconnected, with representation providing

structure, profiling extracting insights, and Anomaly Detection refining, enabling intelligent, scalable, and secure DCN operations.

VII. COMPARATIVE ANALYSIS OF TRADITIONAL AND AI-BASED NETWORKING TECHNIQUES

DCNs have evolved significantly, shifting from rule-based static techniques to adaptive AI-driven solutions. Although traditional methods have provided stability and efficiency, AI/ML-based solutions offer higher adaptability, predictive capabilities, and automation. This section compares both approaches and highlights their advantages, limitations, and emerging trends.

A. CONVENTIONAL TECHNIQUES IN DCN

Traditional DCN methods (N=1371, 84%) rely on predefined rules, heuristics, and static models to optimize network operations. These techniques have been widely used for various network functions, but often lack the flexibility needed to adapt to real-time network fluctuations.

Optical networking commonly employs fixed-wavelength assignment strategies to ensure stable transmission paths. However, these methods fail to adjust in real time based on link conditions, making them inefficient in adaptive networks [168]. Similarly, congestion control mechanisms, such as TCP Reno and TCP Cubic, react only after congestion occurs, leading to scalability issues in large DCN infrastructures [169].

To distribute traffic, scheduling and queue management algorithms such as round-robin scheduling and weighted fair queuing allocate resources fairly but lack responsiveness to sudden traffic spikes [122], [123]. Similarly, rule-based SDN flow management directs traffic through predefined paths, providing control but lacking real-time adaptability [152], [170].

In terms of fault tolerance, failover- and redundancy-based techniques ensure network stability, but introduce additional overhead and infrastructure costs [124], [125]. Likewise, static power management strategies in cloud environments optimize energy consumption under stable conditions but struggle with dynamic workloads, leading to inefficient resource utilization [122].

Traditional methods provide stability, low computational costs, and ease of implementation. However, their reliance on static representation for network structuring, rule-based profiling for monitoring traffic, and signature-based Anomaly Detection makes them less adaptable to real-time fluctuations and large-scale automation, reducing their effectiveness in dynamic environments.

B. AI AND ML DRIVEN APPROACHES IN DCN

AI/ML-based techniques (N=265, 16%) transformed DCN management by enabling real-time decision making, predictive analysis, and self-learning capabilities. Unlike static rule-based approaches, AI-driven solutions dynamically adapt to network conditions, predict congestion, and optimize resource allocation.

In high-speed optical networking, AI models enhance transmission reliability by optimizing signal modulation and predicting link failures [171]. Similarly, AI-powered congestion control enables proactive traffic flow management, thereby preventing bottlenecks before they occur [172]. AI models leverage Representation to structure network data efficiently.

AI/ML also improves real-time scheduling and load balancing, dynamically adjusting network traffic, and prioritizing tasks based on real-time data and network conditions. Deep Deterministic Policy Gradient (DDPG) and Deep Reinforcement Learning algorithms optimize link weights and traffic patterns, ensuring efficient load distribution [173]. Additionally, AI-powered SDN controllers continuously learn from traffic patterns, adjusting routing policies using profiling insights to enhance dynamic load balancing, congestion prediction, and intelligent network traffic management [174].

Furthermore, ML models such as Artificial Neural Networks (ANN), Graph Neural Networks (GNN), and Support Vector Machines (SVM) have been employed to analyze historical traffic patterns and predict traffic surges, enabling proactive resource allocation [175]. Additionally, deep learning architectures such as Residual Networks (ResNet) allow AI-driven systems to estimate network link occupancy in real time, enhancing routing decisions and overall efficiency [176].

Another key advantage of AI-driven networking is fault tolerance and predictive maintenance. AI models detect potential failures before they occur, thereby minimizing the downtime and operational overhead [177]. AI also optimizes energy-efficient networking using reinforcement learning models to predict workload fluctuations and intelligently adjust server utilization, leading to optimized power consumption [176]. These techniques rely on Anomaly Detection to continuously monitor network behavior, detect performance degradation, and mitigate security threats.

By integrating AI/ML across multiple aspects of network operations, modern DCNs are shifting from reactive rule-based frameworks to proactive intelligent systems.

C. AI VS. TRADITIONAL METHODS IN DCN: KEY DIFFERENCES AND EMERGING TRENDS

Traditional networking techniques in DCNs rely on rule-based algorithms, static configurations, and reactive adjustments, making them stable and computationally efficient, but limited in adaptability and scalability. In contrast, AI/ML-driven approaches introduce dynamic decision-making, predictive analytics, and automation, enabling networks to anticipate congestion, optimize traffic flow, and adjust resources in real time.

Traditional methods often require manual intervention and predefined thresholds, whereas AI models continuously learn

from data, making proactive optimizations without human input. In fault tolerance, conventional networks depend on redundancy and failover mechanisms, increasing infrastructure costs, whereas AI-based predictive maintenance identifies failures before they occur, reducing downtime. Similarly, in terms of energy efficiency, traditional networks rely on fixed threshold-based power management, often leading to resource inefficiencies, whereas AI dynamically adjusts power consumption based on real-time workload demands.

Despite AI's advantages, it requires high computational power, extensive datasets, and ongoing optimization, making traditional methods more interpretable, lighter, and simpler to implement in resource-constrained environments. As AI technologies advance, hybrid approaches that combine AI adaptability with the stability of traditional methods are expected to shape the future of autonomous self-optimizing DCNs.

Discussion on RQ4

How do AI/ML techniques leverage network concepts to address DCN challenges, and how do they compare to traditional approaches?

AI/ML enhances DCNs by leveraging key network concepts—Representation, Profiling, and Anomaly Detection—to enable real-time adaptation, predictive optimization, and automation. Representation structures network data for AI-driven traffic management, congestion control, and routing optimization. Profiling helps analyze traffic patterns, allocate resources efficiently, and predict failures. Anomaly Detection strengthens fault tolerance and security by identifying irregularities and mitigating threats in real time.

Unlike static and rule-based methods, AI continuously learns and adjusts, optimizing the performance dynamically. Although AI improves scalability and efficiency, it requires high computational power, making traditional methods more efficient in constrained environments. Future DCNs will integrate hybrid models, balancing AI's adaptability with the stability of traditional approaches for enhanced automation and reliability

VIII. CHALLENGES IN IMPLEMENTING AI/ML FOR DCN OPTIMIZATION AND MANAGEMENT

The implementation of AI/ML solutions follows a structured life cycle comprising of three key phases: design, development, and deployment [178]. In the design phase, AI/ML systems are conceptualized, risks are assessed, and data requirements are identified. The development phase focuses on model selection, training, and optimization to ensure scalability and efficiency. Finally, the deployment phase involves integrating AI/ML models into real-world environments, where performance, latency, and resource constraints must be managed.

As discussed previously, there are several benefits to integrating AI/ML-based solutions for the optimization and management of DCNs; however, key challenges arise that need to be addressed. These challenges span the AI life cycle, including data acquisition limitations, security and privacy concerns, model adaptability, real-time inference constraints, and resource allocation inefficiencies. Figure 4 illustrates the challenges faced in each phase of the lifecycle. The following sections categorize these challenges within the AI life cycle, providing a structured analysis of the barriers to AI-driven DCN optimization.

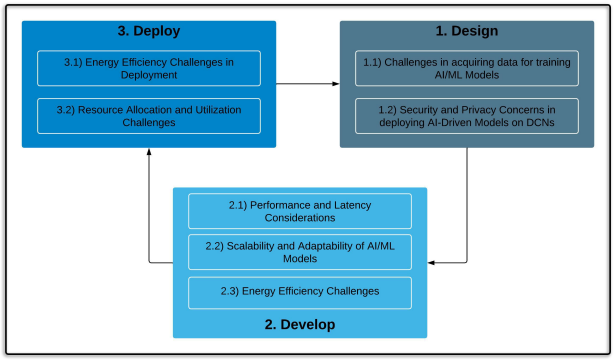


FIGURE 4. The three phases of the AI/ML life cycle and their associated challenges in DCN integration. The design phase struggles with data acquisition, privacy, and security risks, thereby affecting AI feasibility. The development phase faces real-time inference, scalability, and energy efficiency challenges, requiring optimization techniques such as model compression. The deployment phase must manage the resource allocation and high energy consumption to ensure efficient real-time processing. The iterative flow between phases reflects how deployment insights refine design and development strategies.

A. CHALLENGES IN THE DESIGN PHASE

The design phase of AI/ML integration in DCNs involves conceptualizing models, assessing risks, and defining data requirements. Key challenges include limited access to high-quality training data, ensuring privacy and security, and protecting models against adversarial attacks. In the following subsection, we discuss these challenges in detail.

1) CHALLENGES IN ACQUIRING DATA FOR TRAINING AI/ML MODELS

AI models require datasets to be trained, and high-quality datasets are rare, especially in network-related domains where data collection is sometimes limited by privacy concerns, storage constraints, and access restrictions. Many datasets are either proprietary or require many permissions for access, which makes it challenging for researchers to obtain diverse and representative data for training AI models. This is evident in Figure 5 which illustrates the number of datasets published every year. Real-world datasets bring several challenges, including data availability, complexity, and preprocessing requirements. Some datasets, such as the InSDN dataset, collect specific network settings, thereby restricting more general applicability [179]. Large-scale datasets require significant computational resources for training [180]. Another rising concern is privacy

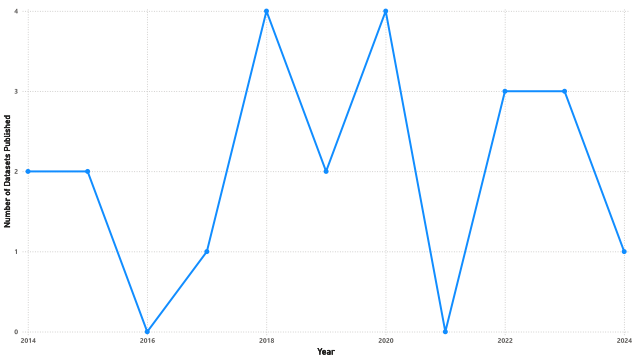


FIGURE 5. Number of datasets on DCN research proposed every year in the past decade.

constraints, which further limit access to sensitive network data, thereby complicating dataset sharing and model reproducibility [181]. Although valuable for cybersecurity research, incident detection datasets frequently suffer from bias and incompleteness [182]. Additionally, real-world network traffic traces require significant preprocessing to manage noise and inconsistencies [183].

Synthetic datasets provide a substitute but have certain natural limitations. Many depend on presumptions regarding traffic models that may not entirely reflect actual behavior [184]. Simulated datasets such as those based on statistical distributions can misrepresent real traffic patterns [185]. Similarly, latency and congestion datasets assume ideal network conditions, thereby overlooking unpredictable network variables [186]. While synthetic data help address privacy concerns, its effectiveness depends on how accurately it mimics actual network behavior [187].

In summary, both synthetic and real-world datasets have difficulties that interfere with the training of AI/ML models. Dealing with these constraints requires better data-sharing systems, updated simulation methods, and hybrid approaches that combine synthetic and real data to improve model adaptability.

2) SECURITY AND PRIVACY CONCERNS IN DEPLOYING AI-DRIVEN MODELS ON DCNS

AI-based network optimization relies on vast amounts of traffic and performance data, which are often collected from multiple sources within a DCN. Ensuring compliance with privacy regulations while maintaining the utility of data is a significant challenge. Techniques such as differential privacy and federated learning can help reduce exposure to sensitive data but introduce trade-offs in model accuracy and convergence speed [188], [189]. ML models used in DCNs are susceptible to adversarial attacks, including data poisoning and model evasion techniques. Attackers can manipulate training data or inference inputs to degrade the model performance, leading to incorrect network optimization decisions. Ensuring robust adversarial training and anomaly detection mechanisms is crucial for mitigating these security threats [190], [191].

B. CHALLENGES IN THE DEVELOPMENT PHASE

The development phase of AI/ML in DCNs focuses on model selection, training, and optimization for scalability and efficiency. The major challenges include real-time inference constraints, high computational demands, and model adaptability to dynamic network conditions. Solutions such as model compression, edge processing, and reinforcement learning help mitigate these issues but often come with trade-offs in performance and resource utilization. In the following subsections, we discuss these challenges in detail.

1) PERFORMANCE AND LATENCY CONSIDERATIONS

Many AI/ML techniques require real-time or near-real-time inference to make network management decisions. However, deep learning models, particularly those used in predictive analytics, can introduce significant inference latency. The challenge lies in deploying lightweight and efficient AI models that are capable of making rapid decisions without compromising accuracy. Techniques such as knowledge distillation and quantization are being explored to reduce inference overhead while maintaining model performance [192], [193]. AI/ML models require continuous data ingestion, processing, and inference, adding overhead to existing DCN operations. Data preprocessing pipelines must be carefully designed to avoid excessive computational costs. Furthermore, the need for distributed AI processing across network nodes introduces synchronization challenges, which can affect the overall performance of AI-driven optimization strategies [194], [195].

2) SCALABILITY AND ADAPTABILITY OF AI/ML MODELS

The deployment of AI/ML models for the optimization and management of DCNs raises significant concerns regarding scalability and adaptability. Given that DCNs are expansive network infrastructures that generate substantial volumes of network telemetry data, the capacity of these models to process and analyze such extensive datasets in real-world prediction scenarios remains highly uncertain. Furthermore, the training of these models requires considerable computational resources, and in some cases, even the inference phase imposes significant resource requirements. This necessitates the adoption of strategies such as model compression and edge processing techniques. However, these approaches may compromise the overall performance of the models, thereby presenting a critical trade-off that must be carefully addressed [195], [196], [197]. DCNs are dynamic environments. These complex networks experience continuous fluctuations in their traffic patterns and workload distributions. AI/ML models must dynamically adapt to these changes, but retraining models in real time presents challenges in terms of computational overhead and convergence time. Reinforcement learning and online learning approaches have been explored to address this issue; however, achieving stable performance while continuously updating the models remains a critical challenge. Furthermore, the

need for continuous data collection and preprocessing for real-time learning adds additional overhead, complicating deployment and resource management [194], [195], [198].

3) ENERGY EFFICIENCY CHALLENGES

The integration of AI/ML models for DCN optimization increases the overall power consumption, particularly during the model training and inference phases. Energy-efficient AI solutions must balance accuracy and computational cost, requiring novel approaches such as lightweight neural networks, hardware accelerators, and adaptive power management strategies. Future research on green AI techniques aims to minimize the carbon footprint of AI-based DCN management while maintaining optimization effectiveness [192], [199].

C. CHALLENGES IN THE DEPLOYMENT PHASE

The deployment phase of AI/ML in DCNs involves integrating models into real-world environments, while managing performance, latency, and resource constraints. Key challenges include efficient resource allocation, high energy consumption, and fairness in workload distribution. Techniques such as federated learning, cloud-edge hybrid approaches, and adaptive power management aim to optimize deployment, but balancing scalability and efficiency remains a challenge. In the following subsections we discuss these challenges in detail.

1) ENERGY EFFICIENCY CHALLENGES IN DEPLOYMENT

The deployment of AI/ML models in DCNs significantly affects energy consumption because continuous inference and resource management require substantial computational power. Unlike the training phase, where energy usage is concentrated in bursts, deployment demands sustained power for real-time model execution, data processing, and network optimization. High power consumption during inference can strain existing infrastructure, increasing operational costs and environmental impact. To address these challenges, lightweight neural networks, hardware accelerators, and adaptive power management strategies have been proposed. Federated learning and cloud-edge hybrid approaches help distribute workloads efficiently, reducing the burden on centralized DCs. However, optimizing AI deployment for energy efficiency without compromising performance remains a critical challenge that requires further advancement in green AI techniques and intelligent power-aware scheduling mechanisms.

2) RESOURCE ALLOCATION AND UTILIZATION CHALLENGES

AI/ML models have been leveraged to optimize resource allocation in DCNs by predicting network congestion, balancing workloads, and improving task scheduling. However, implementing ML-driven resource allocation strategies requires high-granularity data, real-time inference, and robust feedback loops to avoid performance degradation. Additionally, ML models must ensure fairness in resource allocation, prevent bias in task distribution, and avoid resource contention, which can lead to bottlenecks [198], [200], [201]. Although AI/ML models aim to optimize network performance, they introduce resource consumption challenges. Training and deploying large-scale AI models require significant computational resources, often leading to underutilized hardware during non-peak training phases. Implementing AI-driven workload consolidation strategies, such as federated learning or cloud-edge hybrid approaches, can mitigate inefficiencies. However, ensuring the efficient deployment of AI models without increasing power consumption or straining existing network infrastructure remains a concern [191], [192], [201].

Discussion on RQ5

What are the main challenges in designing, developing and deploying AI/ML-based solutions for the optimization and management of DCN?

The primary challenges in DCN research can be categorized into three key areas:

- **Design Challenges**:
  Data scarcity, privacy concerns, and adversarial risks.
- **Development Challenges**:
  Real-time inference constraints, high computational demands, and model adaptability. Solutions such as *model compression* and *edge processing* help but introduce trade-offs.
- **Deployment Challenges**:
  Resource allocation, energy consumption, and fairness concerns are addressed through *federated learning* and *cloud-edge strategies*.

Despite advancements, ensuring efficient and scalable AI-driven DCN solutions requires further exploration of optimization, security, and sustainable deployment strategies.

IX. DISCUSSION

Research on DCNs has gradually grown over the last ten years, illustrating a significant rise in studies on DCN optimization and management. Nonetheless, DCNs remain understudied compared to other fields of DC research. This difference emphasizes the importance of paying more attention to DCNs to ensure that they change in line with developments in other critical DC technologies, storage systems, and cloud computing. Despite this slower pace, researchers have achieved significant progress in addressing issues in network efficiency, reducing latency, enhancing load balancing, and strengthening overall system resilience.

Covering congestion control, traffic engineering, load balancing, SDN, and fault tolerance, this scoping review outlines the major themes of DCN research. These topics show several opportunities and challenges in this field, as well as the collective effort to improve DCN structures and operations. The study also explores how AI and ML are changing network management to provide intelligent automation, traffic flow optimization, system failure prediction, and

traffic flow optimization. In addition, security systems driven by AI and ML are increasingly helping to find and reduce cyber threats, thereby improving the resilience of modern DCNs.

To answer the core research questions proposed, this review highlights major advancements in DCN optimization. AI-powered congestion control, SDN-based traffic strategies, and adaptive load balancing are among the many solutions being actively explored to improve efficiency. Furthermore, emerging AI and ML applications in predictive maintenance and energy-efficient resource management underscore the importance of automation in DCNs. Hybrid architectures that integrate optical and wireless technologies have been developed, resulting in faster data transmission and reduced latency. By combining these innovations, this review provides a holistic perspective on the transformation of DCNs in response to increasing computational and network demands.

Another crucial aspect of DCN research revolves around modeling and representation, profiling, and anomaly detection, which serve as fundamental networking principles for improving efficiency and security. Modeling and representation provide structured frameworks for analyzing network traffic and optimizing routing, thereby allowing for more adaptive and scalable architectures. Profiling techniques enable a deeper understanding of traffic patterns, network behaviors, and resource utilization, which in turn aids in the dynamic allocation of workloads and improved congestion management. Anomaly detection plays a vital role in ensuring network resilience by identifying security threats, performance bottlenecks, and points of failure before they cause disruptions. Applying these techniques holistically could help overcome many existing DCN challenges, paving the way for a more robust, intelligent, and self-optimizing network infrastructure.

By addressing RQ1–RQ5, our study demonstrates that (1) telemetry-driven profiling (RQ1) provides a reliable baseline for normal network behavior; (2) feature-aware forecasting models (RQ2) significantly improve short-term prediction accuracy; (3) graph-based anomaly detection (RQ3) effectively integrates structural and temporal data to reduce false positives; (4) causal diagnosis via event graphs (RQ4) offers interpretable insights into anomaly origins; and (5) the proposed framework's scalability and modularity (RQ5) support seamless deployment in large-scale data center environments. Collectively, these findings confirm the efficacy of an AI-driven, multistage approach to proactive DCN monitoring and pave the way for future enhancements in real-time network management.

One of the key contributions of this review is its structured analysis of DCN research across multiple themes, providing a well-rounded view of the current progress and areas that still require attention. The scoping review methodology, guided by PRISMA-ScR, ensures a thorough and methodological examination of the existing research. Additionally, the introduction of an LLM-based study screening and topic modeling approach showcases how automation can enhance literature reviews in large-scale research areas. These insights serve as valuable resources for both researchers and industry practitioners. The discussion of AI and ML integration within DCNs further deepens the understanding of both the advantages and challenges of deploying AI-based solutions at scale.

This review contributes to DCN research by providing a structured synthesis of key themes, challenges, and emerging trends across the field. By identifying gaps in areas such as real-time anomaly detection, adaptive traffic control, and energy-efficient network management, this review highlights actionable research directions for optimizing DCN performance. In addition, the scoping methodology adopted demonstrates how automation can enhance large-scale literature mapping, supporting more systematic and informed advancements in DCN design and operation.

X. FUTURE TRENDS IN DCN RESEARCH

The evolution of DCNs has been shaped by growing demands for automation, edge intelligence, sustainability, and security. As data centers increasingly support cloud-native, AI-driven, and latency-sensitive services, several promising avenues for future research have emerged.

The integration of AI in Radio Access Networks (AI-RAN) and Multi-access Edge Computing (AI-MEC) is set to revolutionize network efficiency and responsiveness. AI-RAN effectively utilizes machine learning algorithms to optimize resource allocation and enhance user experience by predicting traffic patterns and dynamically adjusting network parameters [202]. Although AI-MEC also uses edge-computing capabilities to process data closer to the source, thus reducing latency and increasing operational efficiency, its specific impact on real-time applications must be explored through further research [203].

The concept of dark network operation centers (DNOCs), characterized by minimal to no human intervention in data management and operations, is becoming a significant trend in the evolution of data centers. This transformation is primarily driven by the increasing need for operational efficiency and energy affordability within the context of cloud computing and extensive digital infrastructure. The integration of advanced automation technologies plays a crucial role in this shift, allowing for autonomous monitoring and control of data center operations. For instance, new edge computing platforms facilitate intelligent operational monitoring using wireless and built-in sensors, which optimizes workload distribution, thereby enhancing energy efficiency and operational reliability [204]. Moreover, innovative architectures utilizing bidirectional DC-DC converters are important for ensuring safety and stability in power supply systems within DNOCs, especially given the growing operational demands for power management in geo-distributed data centers that support global cloud services [205], [206]. Another essential element influencing this trend is ongoing research on resource management strategies aimed at minimizing costs and improving performance across various operational

environments [207]. The emergence of autonomous systems in data centers signifies broader technological advancement, where reduced human oversight can optimize efficiency and decrease operational errors while enhancing scalability. This shift towards a more sustainable and efficient technological infrastructure is critical for managing increasing workloads and energy demands [208], [209].

Sustainability has become a primary focus for future data center network research, with significant efforts directed towards reducing the carbon footprint and enhancing energy efficiency. Strategies such as transitioning to renewable energy sources, optimizing cooling systems, and implementing circular economy practices are crucial for minimizing environmental impacts [210], [211]. Studies have indicated that substantive energy savings can be achieved through innovative management practices, including energy conservation strategies and AI-assisted resource allocation techniques [212], [213], [214].

The integration of quantum computing into data center security frameworks presents an unprecedented opportunity to enhance data protection strategies. Quantum cryptography can offer significantly higher levels of security than classical methods by enabling secure communication channels that are theoretically immune to eavesdropping [215]. As data centers store vast amounts of sensitive information, the application of these advanced security measures will become increasingly critical for safeguarding against evolving cyber threats [216].

The landscape of DCNs is rapidly evolving, driven by advancements in AI, sustainability imperatives, enhanced security protocols, and a focus on programmable infrastructure. As these trends unfold, research will play a critical role in developing more efficient, resilient, and secure data center operations to meet the demands of a data-driven future. With a continued commitment to innovative approaches, the data center industry can aspire to a more sustainable and intelligent framework.

XI. CONCLUSION

In this scoping review, we mapped the existing body of research and synthesized the prevailing themes, methods, and challenges across the DCN domain. Our goal is to provide a structured overview that highlights key trends and identifies areas where further technology-specific investigations can drive meaningful advancements. Several exciting research directions are being explored. Future efforts should focus on seamlessly integrating AI and ML solutions with real-time network operations to enhance adaptability and predictive capabilities. More comprehensive benchmarking frameworks are required to evaluate AI-driven networking solutions relative to traditional methods. Additionally, sustainability remains a crucial aspect, and research into energy-efficient DCN designs is key to reducing the environmental footprint of large-scale DCs. Other promising areas include quantum networking and blockchains, which can provide enhanced security and decentralized network management. As DCNs continue to advance, interdisciplinary collaboration between networking, ML, and cloud computing will shape next-generation architectures. The growing adoption of edge computing and 5G presents exciting opportunities for optimizing data flow and minimizing latency in distributed environments. Ultimately, AI-driven solutions will likely become the cornerstone of future DCN optimization, enabling networks to operate with greater efficiency, intelligence, and resilience.

Finally, this review presents a comprehensive review of DCN research, shedding light on key innovations, challenges, and future possibilities. By addressing gaps in the existing literature and identifying opportunities for further study, this study contributes to broader efforts to enhance the performance, scalability, and sustainability of DCNs. As technological advancements accelerate, ongoing research and collaboration will be essential in refining current methodologies and driving the next wave of breakthroughs in DCN optimization. The future of DCNs is promising, and with AI continuing to lead the way, these networks will become more autonomous, adaptive, and indispensable for supporting the ever-expanding digital landscape.

APPENDIX

LIST OF ABBREVIATIONS

Abbreviation	Full Term
AI	Artificial Intelligence
API	Application Programming Interface
BGP	Border Gateway Protocol
CPU	Central Processing Unit
DCN	Data Center Network
DNS	Domain Name System
FPGA	Field-Programmable Gate Array
GPU	Graphics Processing Unit
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IoT	Internet of Things
LAN	Local Area Network
ML	Machine Learning
MPLS	Multiprotocol Label Switching
NFV	Network Function Virtualization
NIC	Network Interface Card
OSPF	Open Shortest Path First
PUE	Power Usage Effectiveness
QoS	Quality of Service
RTT	Round-Trip Time
SDN	Software-Defined Networking
SLA	Service Level Agreement
SSD	Solid State Drive
TCO	Total Cost of Ownership
TCP	Transmission Control Protocol
VLAN	Virtual Local Area Network
VM	Virtual Machine
WAN	Wide Area Network

ACKNOWLEDGMENT

(Kogul Srikandabala and Kirishnni Prabagar contributed equally to this work.)

REFERENCES

[1] C. Wu and R. Buyya, *Cloud Data Centers and Cost Modeling*. San Mateo, CA, USA: Morgan Kaufmann, Mar. 2015.

[2] W. Xia, P. Zhao, Y. Wen, and H. Xie, "A survey on data center networking (DCN): Infrastructure and operations," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 1, pp. 640–656, 1st Quart., 2017.

[3] R. R. Schaller, "Moore's law: Past, present and future," *IEEE Spectrum*, vol. 34, no. 6, pp. 52–59, Jun. 1997.

[4] R. McGeer, P. Mahadevan, and S. Banerjee, "On the complexity of power minimization schemes in data center networks," in *Proc. IEEE Global Telecommun. Conf. GLOBECOM*, Dec. 2010, pp. 1–5.

[5] L. Popa, S. Ratnasamy, G. Iannaccone, A. Krishnamurthy, and I. Stoica, "A cost comparison of datacenter network architectures," in *Proc. 6th Int. Conf.*, Nov. 2010, pp. 1–12.

[6] C. E. Rothenberg, "Re-architected cloud data center networks and their impact on the future Internet," in *New Network Architectures*. Cham, Switzerland: Springer, 2010, pp. 179–187.

[7] L. Schares, D. M. Kuchta, and A. F. Benner, "Optics in future data center networks," in *Proc. 18th IEEE Symp. High Perform. Interconnects*, Aug. 2010, pp. 104–108.

[8] Y. Shang, D. Li, and M. Xu, "A comparison study of energy proportionality of data center network architectures," in *Proc. 32nd Int. Conf. Distrib. Comput. Syst. Workshops*, Jun. 2012, pp. 1–7.

[9] K. Wu, J. Xiao, and L. M. Ni, "Rethinking the architecture design of data center networks," *Frontiers Comput. Sci.*, vol. 6, no. 5, pp. 596–603, Sep. 2012.

[10] R. P. Tahiliani, M. P. Tahiliani, and K. C. Sekaran, "TCP variants for data center networks: A comparative study," in *Proc. Int. Symp. Cloud Services Comput.*, Dec. 2012, pp. 57–62.

[11] C. Kachris and I. Tomkos, "The rise of optical interconnects in data centre networks," in *Proc. 14th Int. Conf. Transparent Opt. Netw. (ICTON)*, Jul. 2012, pp. 1–4.

[12] L. Xu, A. Singh, and Y. Zhang, "Optically interconnected data center networks," in *Proc. Opt. Fiber Commun. Conf.*, 2012, pp. 1–3.

[13] X.-L. Wei, M. Chen, J.-H. Fan, G.-M. Zhang, and Z.-Y. Lu, "Architecture of the data center network: Architecture of the data center network," *J. Softw.*, vol. 24, no. 2, pp. 295–316, Dec. 2013.

[14] N. Farrington and A. Andreyev, "Facebook's data center network architecture," in *Proc. Opt. Interconnects Conf.*, May 2013, pp. 49–50.

[15] J. Zhang, F. Ren, and C. Lin, "Survey on transport control in data center networks," *IEEE Netw.*, vol. 27, no. 4, pp. 22–26, Jul. 2013.

[16] J. Perelló et al., "All-optical packet/circuit switching-based data center network for enhanced scalability, latency, and throughput," *IEEE Netw.*, vol. 27, no. 6, pp. 14–22, Nov. 2013.

[17] C. Kachris and I. Tomkos, "Power consumption evaluation of all-optical data center networks," *Cluster Comput.*, vol. 16, no. 3, pp. 611–623, Aug. 2012.

[18] K. Bilal, S. U. Khan, and A. Y. Zomaya, "Green data center networks: Challenges and opportunities," in *Proc. 11th Int. Conf. Frontiers Inf. Technol.*, Dec. 2013, pp. 229–234.

[19] P. P. Jiang, C. Q. Yu, and Y. H. Peng, "Energy-saving in optical data center networks," *Appl. Mech. Mater.*, vols. 411–414, pp. 634–637, Sep. 2013.

[20] M. F. Bari, R. Boutaba, R. Esteves, L. Z. Granville, M. Podlesny, M. G. Rabbani, Q. Zhang, and M. F. Zhani, "Data center network virtualization: A survey," *IEEE Commun. Surveys Tuts.*, vol. 15, no. 2, pp. 909–928, 2nd Quart., 2013.

[21] Y. Cai, Y. Yan, Z. Zhang, and Y. Yang, "Survey on converged data center networks with DCB and FCoE: Standards and protocols," *IEEE Netw.*, vol. 27, no. 4, pp. 27–32, Jul. 2013.

[22] A. Hendel, "Large- scale data center networks," in *Proc. Opt. Fiber Commun. Conf.*, 2014, pp. Th4J.1–Th4J.1.

[23] F. Yao, J. Wu, G. Venkataramani, and S. Subramaniam, "A comparative analysis of data center network architectures," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Jun. 2014, pp. 3106–3111.

[24] Y. Ren, Y. Zhao, P. Liu, K. Dou, and J. Li, "A survey on TCP incast in data center networks," *Int. J. Commun. Syst.*, vol. 27, no. 8, pp. 1160–1172, Jul. 2012.

[25] H. X. Wu, X. L. Yang, and M. Zhang, "Congestion control in data center networks: A survey and new perspectives," *Appl. Mech. Mater.*, vols. 462–463, pp. 1028–1035, Nov. 2013.

[26] K.-i. Kitayama, Y.-C. Huang, Y. Yoshida, R. Takahashi, and M. Hayashitani, "Optical packet and path switching intra-data center network: Enabling technologies and network performance with intelligent flow control," in *Proc. Eur. Conf. Opt. Commun. (ECOC)*, Sep. 2014, pp. 1–3.

[27] M. Fayyaz and K. Aziz, "Classification of optical interconnects in data center networks," in *Proc. 12th Int. Conf. Frontiers Inf. Technol.*, Dec. 2014, pp. 61–66.

[28] N. Calabretta, W. Miao, S. D. Lucente, J. Luo, and H. J. S. Dorren, "Scalable and low latency optical packet switching architectures for high performance data center networks," in *Adv. Photon. Commun.*, 2014.

[29] K. Kitayama, Y. Huang, Y. Yoshida, R. Takahashi, and M. Hayashitani, "OPS/OCS intra-data center network with intelligent flow control," in *Adv. Photon. Commun.*, 2014.

[30] H. Qi, M. Shiraz, J.-Y. Liu, A. Gani, Z. A. Rahman, and T. A. Altameem, "Data center network architecture in cloud computing: Review, taxonomy, and open research issues," *J. Zhejiang Univ. Sci. C*, vol. 15, no. 9, pp. 776–793, Sep. 2014.

[31] A. Hammadi and L. Mhamdi, "A survey on architectures and energy efficiency in data center networks," *Comput. Commun.*, vol. 40, pp. 1–21, Mar. 2014.

[32] K. Bilal, S. U. R. Malik, O. Khalid, A. Hameed, E. Alvarez, V. Wijaysekara, R. Irfan, S. Shrestha, D. Dwivedy, M. Ali, U. S. Khan, A. Abbas, N. Jalil, and S. U. Khan, "A taxonomy and survey on green data center networks," *Future Gener. Comput. Syst.*, vol. 36, pp. 189–208, Jul. 2014.

[33] B. Maggs, "A universal approach to data center network design," in *Proc. 26th ACM Symp. Parallelism Algorithms Archit.*, Jun. 2014, p. 246.

[34] A. Lonare and V. Gulhane, "Addressing agility and improving load balance in fat-tree data center network—A review," in *Proc. 2nd Int. Conf. Electron. Commun. Syst. (ICECS)*, Feb. 2015, pp. 965–971.

[35] M. Fayyaz, K. Aziz, and G. Mujtaba, "Blocking probability in optical interconnects in data center networks," *Photonic Netw. Commun.*, vol. 30, no. 2, pp. 204–222, May 2015.

[36] W. Hou, L. Guo, Y. Liu, C. Yu, and Y. Zong, "Resource management and control in converged optical data center networks: Survey and enabling technologies," *Comput. Netw.*, vol. 88, pp. 121–135, Sep. 2015.

[37] R. Takahashi, T. Segawa, S. Ibrahim, T. Nakahara, H. Ishikawa, A. Hiramatsu, Y.-C. Huang, and K.-i. Kitayama, "Torus data center network with smart flow control enabled by hybrid optoelectronic routers [Invited]," *J. Opt. Commun. Netw.*, vol. 7, no. 12, pp. B141–B152, Dec. 2015.

[38] C. Kachris and I. Tomkos, "A roadmap on optical interconnects in data centre networks," in *Proc. 17th Int. Conf. Transparent Opt. Netw. (ICTON)*, Jul. 2015, pp. 1–3.

[39] K.-I. Kitayama, Y.-C. Huang, Y. Yoshida, R. Takahashi, T. Segawa, S. Ibrahim, T. Nakahara, Y. Suzaki, M. Hayashitani, Y. Hasegawa, Y. Mizukoshi, and A. Hiramatsu, "Torus-topology data center network based on optical Packet/Agile circuit switching with intelligent flow management," *J. Lightw. Technol.*, vol. 33, no. 5, pp. 1063–1071, Mar. 1, 2015.

[40] R. Rojas-Cessa, Y. Kaymak, and Z. Dong, "Schemes for fast transmission of flows in data center networks," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 3, pp. 1391–1422, 3rd Quart., 2015.

[41] E. Baccour, S. Foufou, R. Hamila, and M. Hamdi, "A survey of wireless data center networks," in *Proc. 49th Annu. Conf. Inf. Sci. Syst. (CISS)*, Mar. 2015, pp. 1–6.

[42] C. Yu, W. Hou, and L. Guo, "A survey on virtual network embedding in optical cloud data center networks," in *Proc. Int. Conf. Softw. Netw. (ICSN)*, May 2016, pp. 1–5.

[43] P. Sreekumari and J.-I. Jung, "Transport protocols for data center networks: A survey of issues, solutions and challenges," *Photonic Netw. Commun.*, vol. 31, no. 1, pp. 112–128, Aug. 2015.

[44] A. Shpiner and E. Zahavi, "Race cars vs. Trailer trucks: Switch buffers sizing vs. Latency trade-offs in data center networks," in *Proc. IEEE 24th Annu. Symp. High-Performance Interconnects (HOTI)*, Aug. 2016, pp. 53–59.

[45] O. Raz, G. Guelbenzu, T. Li, C. Li, W. Miao, F. Yan, H. J. S. Dorren, P. Stabile, and N. Calabretta, "Optical solutions for the challenges of mega-size data center networks," in *Proc. Opt. Fiber Commun. Conf. Exhib. (OFC)*, Mar. 2016, pp. 1–3.
[46] M. Fayyaz, K. Aziz, and G. Mujtaba, "Performance analysis of optical interconnects' architectures for data center networks: Do you have a subtitle? If so, write it here," *Cluster Comput.*, vol. 19, no. 3, pp. 1139–1161, Jul. 2016.
[47] T. Chen, X. Gao, and G. Chen, "The features, hardware, and architectures of data center networks: A survey," *J. Parallel Distrib. Comput.*, vol. 96, pp. 45–74, Oct. 2016.
[48] V. D. Chakravarthy and V. Nagarajan, "Techniques for optimizing power utilization in data center network architectures: A survey report," *Indian J. Sci. Technol.*, vol. 9, no. 37, pp. 1–6, Oct. 2016.
[49] T. Hafeez, N. Ahmed, B. Ahmed, and A. W. Malik, "Detection and mitigation of congestion in SDN enabled data center networks: A survey," *IEEE Access*, vol. 6, pp. 1730–1740, 2018.
[50] G. C. Sankaran and K. M. Sivalingam, "A survey of hybrid optical data center network architectures," *Photonic Netw. Commun.*, vol. 33, no. 2, pp. 87–101, Jul. 2016.
[51] S. Yan, "Recent research in intra/inter-data center network communication," in *Proc. Asia Commun. Photon. Conf. (ACP)*, Nov. 2017, pp. 1–12.
[52] D. Zhang, H. Guo, G. Chen, Y. Zhu, H. Yu, J. Wang, and J. Wu, "Analysis and experimental demonstration of an optical switching enabled scalable data center network architecture," *Opt. Switching Netw.*, vol. 23, pp. 205–214, Jan. 2017.
[53] Y. Tang, H. Guo, X. Li, and J. Wu, "Experimental demonstration of efficient topology reconstruction in optical switching based data center network," in *Proc. 16th Int. Conf. Opt. Commun. Netw. (ICOCN)*, Aug. 2017, pp. 1–3.
[54] B. Dai, G. Xu, B. Huang, P. Qin, and Y. Xu, "Enabling network innovation in data center networks with software defined networking: A survey," *J. Netw. Comput. Appl.*, vol. 94, pp. 33–49, Sep. 2017.
[55] S. Subramaniam, "Optics in data center networks," *Proc. Adv. Photon. 2017 (IPR, NOMA, Sensors, Netw., SPPCom, PS) OSA*, 2017, Paper NeW2B.2, doi: 10.1364/networks.2017.new2b.2.
[56] Z. Han and L. Yu, "A survey of the BCube data center network topology," in *Proc. IEEE IEEE 4th Int. Conf. Big Data Secur. Cloud (BigDataSecurity) Int. Conf. High Perform. Smart Comput., (HPSC) IEEE Int. Conf. Intell. Data Secur. (IDS)*, May 2018, pp. 229–231.
[57] Z. Han and W. Zhang, "A summary of the BCCC data center network topology," in *Proc. IEEE IEEE 4th Int. Conf. Big Data Secur. Cloud (BigDataSecurity) Int. Conf. High Perform. Smart Comput., (HPSC) IEEE Int. Conf. Intell. Data Secur. (IDS)*, May 2018, pp. 270–272.
[58] Z. Chkirbene, R. Hamila, and S. Foufou, "A survey on data center network topologies," in *Ubiquitous Networking* (Lecture notes in computer science). Cham, Switzerland: Springer, 2018, pp. 143–154.
[59] A. N. Quttoum, "Interconnection structures, management and routing challenges in cloud-service data center networks: A survey," *Int. J. Interact. Mobile Technol. (iJIM)*, vol. 12, no. 1, p. 36, Jan. 2018.
[60] X. Jiang, M. D. Feuer, and M. V. Bnyamin, "Advanced modulation formats for large capacity data center networks," *Proc. SPIE*, vol. 10536, p. 71, May 2018.
[61] K. Wang, Q. Zhou, S. Guo, and J. Luo, "Cluster frameworks for efficient scheduling and resource allocation in data center networks: A survey," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 3560–3580, 2018.
[62] S. Wang, J. Zhang, T. Huang, J. Liu, T. Pan, and Y. Liu, "A survey of coflow scheduling schemes for data center networks," *IEEE Commun. Mag.*, vol. 56, no. 6, pp. 179–185, Jun. 2018.
[63] J. Zhang, F. R. Yu, S. Wang, T. Huang, Z. Liu, and Y. Liu, "Load balancing in data center networks: A survey," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 3, pp. 2324–2352, 3rd Quart., 2018.
[64] S. Huang, D. Dong, and W. Bai, "Congestion control in high-speed lossless data center networks: A survey," *Future Gener. Comput. Syst.*, vol. 89, pp. 360–374, Dec. 2018.
[65] A. Tzanakaki, L. Wosinska, L. Schares, H. Liu, and D. Simeonidou, "Special issue on optical data center networks," *J. Opt. Commun. Netw.*, vol. 10, no. 7, p. ODC1, Jun. 2018.
[66] A. Pilimon, A. M. Kentis, S. Ruepp, and L. Dittmann, "Analysis of traffic engineering capabilities for SDN-based data center networks," in *Proc. 5th Int. Conf. Softw. Defined Syst. (SDS)*, Apr. 2018, pp. 211–216.
[67] R. R. Reyes, S. Sultana, V. V. Pai, and T. Bauschert, "Analysis and evaluation of CAPEX and OPEX in intra-data centre network architectures," in *Proc. IEEE Latin-Amer. Conf. Commun. (LATINCOM)*, Nov. 2019, pp. 1–6.
[68] X. Du, Z. Han, and Y. Huangfu, "A summary of Hamiltonian based on data center network," in *Proc. IEEE 5th Int. Conf. Big Data Secur. Cloud (BigDataSecurity) Int. Conf. High Perform. Smart Comput., (HPSC) IEEE Int. Conf. Intell. Data Secur. (IDS)*, May 2019, pp. 189–191.
[69] M. Klymash, O. Shpur, O. Lavriv, and N. Peleh, "Information security in virtualized data center network," in *Proc. 3rd Int. Conf. Adv. Inf. Commun. Technol. (AICT)*, Jul. 2019, pp. 419–422.
[70] M. Alipio, N. M. Tiglao, F. Bokhari, and S. Khalid, "TCP incast solutions in data center networks: A classification and survey," *J. Netw. Comput. Appl.*, vol. 146, Nov. 2019, Art. no. 102421.
[71] S. Ibrahim and T. Hashimoto, "Novel lambda-rich data center network: From underlying principles to candidate technologies," in *Proc. Opt. Fiber Commun. Conf. (OFC)*, 2019, pp. 1–14.
[72] L. Gonzalez-Naharro, J. Escudero-Sahuquillo, P. J. Garcia, F. J. Quiles, J. Duato, W. Sun, X. Yu, and H. Zheng, "Modeling traffic workloads in data-center network simulation tools," in *Proc. Int. Conf. High Perform. Comput. Simul. (HPCS)*, Jul. 2019, pp. 1036–1042.
[73] Y. Tang, H. Guo, Y. Zhu, T. Yuan, X. Gao, C. Wang, and J. Wu, "Effectively reconfigure the optical circuit switching layer topology in data center network by OCBridge," *J. Lightw. Technol.*, vol. 37, no. 3, pp. 897–908, Feb. 1, 2019.
[74] A. Sharma and R. G. Sangeetha, "Comparative study of optical interconnection architectures in data center networks," *J. Opt. Commun.*, vol. 40, no. 3, pp. 225–238, Jan. 2018.
[75] B. Pavithra, S. Suchitra, S. Sophia, and J. L. George, "SDN based energy efficient cloud data center networks," *Int. J. Innov. Technol. Exploring Eng.*, vol. 8, no. 12, pp. 4250–4256, Oct. 2019.
[76] E. Baccour, S. Foufou, R. Hamila, and A. Erbad, "Green data center networks: A holistic survey and design guidelines," in *Proc. 15th Int. Wireless Commun. Mobile Comput. Conf. (IWCMC)*, Jun. 2019, pp. 1108–1114.
[77] A. S. Hamza, "Recent advances in the design of optical wireless data center networks," *Proc. SPIE*, vol. 10945, p. 20, Feb. 2019.
[78] A. Çelik, B. Shihada, and M. Alouini, "Optical wireless data center networks: Potentials, limitations, and prospects," *Proc. SPIE*, vol. 10945, p. 18, Feb. 2019.
[79] V. Sharma and R. Mishra, "A comprehensive survey on data center network architectures," in *Proc. 8th Int. Conf. Rel., Infocom Technol. Optim. (Trends Future Directions) (ICRITO)*, Jun. 2020, pp. 222–228.
[80] Y.-C. Tsai, T.-C. Hou, and M.-C. Chiu, "Design of TCP congestion control in data center networks based on stable round trip time," in *Proc. 8th Int. Conf. Commun. Broadband Netw.*, Apr. 2020, pp. 40–44.
[81] Z. Guo, S. Liu, and Z.-L. Zhang, "Traffic control for RDMA-enabled data center networks: A survey," *IEEE Syst. J.*, vol. 14, no. 1, pp. 677–688, Mar. 2020.
[82] L. Zhang, J. Chen, E. Agrell, R. Lin, and L. Wosinska, "Enabling technologies for optical data center networks: Spatial division multiplexing," *J. Lightw. Technol.*, vol. 38, no. 1, pp. 18–30, Jan. 1, 2020.
[83] S. Wang, H. Cao, and L. Yang, "A survey of service function chains orchestration in data center networks," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, Dec. 2020, pp. 1–6.
[84] G. S. Kuaban, T. Atmaca, A. Kamli, T. Czachórski, and P. Czekalski, "Performance analysis of packet aggregation mechanisms and their applications in access (E.G., IoT, 4G/5G), core, and data centre networks," *Sensors*, vol. 21, no. 11, p. 3898, Jun. 2021.
[85] T. Sahoo, S. K. Naik, and K. K. Das, "Data center networks: The evolving scenario," in *Smart Innovation, Systems and Technologies*, 2021, pp. 601–610.
[86] J. Jia and G. Liu, "Data center networks: Address scheming, traffic distribution, and load balancing," in *Proc. IEEE Int. Conf. Comput. Sci., Artif. Intell. Electron. Eng. (CSAIEE)*, Aug. 2021, pp. 309–318.
[87] H. Amari, W. Louati, L. Khoukhi, and L. H. Belguith, "TCP incast solutions in data center networks: Survey," in *Advances in Intelligent Systems and Computing*. Cham, Switzerland: Springer, 2021, pp. 535–545.
[88] W. Li, J. Liu, S. Wang, T. Zhang, S. Zou, J. Hu, W. Jiang, and J. Huang, "Survey on traffic management in data center network: From link layer to application layer," *IEEE Access*, vol. 9, pp. 38427–38456, 2021.
[89] Y. Tang, T. Yuan, B. Liu, and C. Xiao, "Effective-flow schedule for optical circuit switching based data center networks: A comprehensive survey," *Comput. Netw.*, vol. 197, Oct. 2021, Art. no. 108321.
[90] H. Dong, A. Munir, H. Tout, and Y. Ganjali, "Next-generation data center network enabled by machine learning: Review, challenges, and opportunities," *IEEE Access*, vol. 9, pp. 136459–136475, 2021.
[91] C. Terzi and I. Korpeoglu, "60 GHz wireless data center networks: A survey," *Comput. Netw.*, vol. 185, Feb. 2021, Art. no. 107730.

[92] T. Niu and H. Li, "A survey of energy-efficient methods in data center networks based on software defined network," in *Proc. IEEE Conf. Telecommun., Opt. Comput. Sci. (TOCS)*, Dec. 2021, pp. 477–483.

[93] R. Tang, "Innovations for data center network optimization," in *Proc. Int. Conf. Comput. Inf. Sci. Artif. Intell. (CISAI)*, Sep. 2021, pp. 610–615.

[94] A. M. Abdelrahman, J. J. P. C. Rodrigues, M. M. E. Mahmoud, K. Saleem, A. K. Das, V. Korotaev, and S. A. Kozlov, "Software-defined networking security for private data center networks and clouds: Vulnerabilities, attacks, countermeasures, and solutions," *Int. J. Commun. Syst.*, vol. 34, no. 4, Dec. 2020, Art. no. e4706.

[95] M. Nooruzzaman and X. Fernando, "Hyperscale data center networks with transparent HyperX architecture," *IEEE Commun. Mag.*, vol. 59, no. 6, pp. 120–125, Jun. 2021.

[96] A. C. Castillo, "Various network topologies and an analysis comparative between fat-tree and BCube for a data center network: An overview," in *Proc. IEEE Cloud Summit*, Oct. 2022, pp. 1–8.

[97] D. Li, G. Cao, L. Xiao, J. Yao, and X. Cao, "Research progress and trend of coflow time-optimal scheduling in data center network," in *Proc. Int. Conf. Artif. Intell. Secur.*, 2022, pp. 560–572.

[98] C. Xie and B. Zhang, "Scaling optical interconnects for hyperscale data center networks," *Proc. IEEE*, vol. 110, no. 11, pp. 1699–1713, Nov. 2022.

[99] B. Li, T. Wang, P. Yang, M. Chen, and M. Hamdi, "Rethinking data center networks: Machine learning enables network intelligence," *J. Commun. Inf. Netw.*, vol. 7, no. 2, pp. 157–169, Jun. 2022.

[100] S. Pachnicke, S. Oettinghaus, S. Calabró, T. Wettlin, N. Stojanovic, and T. Rahman, "Digital signal processing for 1.6 Tb/s+ intra-data center networks," in *Proc. IEEE Photon. Conf. (IPC)*, Nov. 2023, pp. 1–2.

[101] W. Xue, Z. Han, and X. Du, "Review of new data center network structure," in *Proc. 25th Int. Conf. Adv. Commun. Technol. (ICACT)*, Feb. 2023, pp. 232–235.

[102] M. Zhao, Z. Han, and X. Du, "A survey of data center network topology structure," in *Proc. 25th Int. Conf. Adv. Commun. Technol. (ICACT)*, Feb. 2023, pp. 303–309.

[103] C. Nandhini and G. P. Gupta, "Exploration and evaluation of congestion control algorithms for data center networks," *Social Netw. Comput. Sci.*, vol. 4, no. 5, Jun. 2023, Art. no. 509.

[104] R. Ji, L. Ji, K. Gu, J. Wu, and G. Lei, "A survey of multipath transport mechanism in data center networks," in *Proc. Mobile Networks Manag.*, in Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering. Cham, Switzerland: Springer, 2023, pp. 373–385.

[105] A. Feng, D. Dong, F. Lei, J. Ma, E. Yu, and R. Wang, "In-network aggregation for data center networks: A survey," *Comput. Commun.*, vol. 198, pp. 63–76, Jan. 2023.

[106] S. Hu, X. Zhang, C. Li, H. Yin, X. Li, and X. Xin, "Latest advances in VCSEL technology for next-generation data center network [Invited]," *Chin. Opt. Lett.*, vol. 22, no. 11, 2024, Art. no. 111401.

[107] Y. Liu, T. Yu, Q. Meng, and Q. Liu, "Flow optimization strategies in data center networks: A survey," *J. Netw. Comput. Appl.*, vol. 226, Jun. 2024, Art. no. 103883.

[108] A. C. Tricco et al., "PRISMA extension for scoping reviews (PRISMA-ScR): Checklist and explanation," *Ann. Internal Med.*, vol. 169, no. 7, pp. 467–473, Oct. 2018.

[109] F. Dennstädt, J. Zink, P. M. Putora, J. Hastings, and N. Cihoric, "Title and abstract screening for literature reviews using large language models: An exploratory study in the biomedical domain," *Systematic Rev.*, vol. 13, no. 1, Jun. 2024, Art. no. 158.

[110] M. Mostafapour, J. H. Fortier, K. Pacheco, H. Murray, and G. Garber, "Evaluating literature reviews conducted by humans versus ChatGPT: Comparative study," *JMIR AI*, vol. 3, Aug. 2024, Art. no. e56537.

[111] B. Hasan, S. Saadi, N. S. Rajjoub, M. Hegazi, M. Al-Kordi, F. Fleti, M. Farah, I. B. Riaz, I. Banerjee, Z. Wang, and M. H. Murad, "Integrating large language models in systematic reviews: A framework and case study using ROBINS-I for risk of bias assessment," *BMJ Evidence-Based Med.*, vol. 29, no. 6, pp. 394–398, Dec. 2024.

[112] M. Chelli, J. Descamps, V. Lavoué, C. Trojani, M. Azar, M. Deckert, J.-L. Raynier, G. Clowez, P. Boileau, and C. Ruetsch-Chelli, "Hallucination rates and reference accuracy of ChatGPT and bard for systematic reviews: Comparative analysis," *J. Med. Internet Res.*, vol. 26, May 2024, Art. no. e53164.

[113] K. Wang, T. Song, Y. Wang, C. Fang, J. He, A. Nirmalathas, C. Lim, E. Wong, and S. Kandeepan, "Evolution of short-range optical wireless communications," *J. Lightw. Technol.*, vol. 41, no. 4, pp. 1019–1040, Feb. 15, 2023.

[114] J. Wei, T. Rahman, S. Calabrò, N. Stojanovic, L. Zhang, C. Xie, Z. Ye, and M. Kuschnerov, "Experimental demonstration of advanced modulation formats for data center networks on 200 Gb/s lane rate IMDD links," *Opt. Exp.*, vol. 28, no. 23, p. 35240, Nov. 2020.

[115] C. Browning, Q. Cheng, N. C. Abrams, M. Ruffini, L. Y. Dai, L. P. Barry, and K. Bergman, "A silicon photonic switching platform for flexible converged centralized-radio access networking," *J. Lightw. Technol.*, vol. 38, no. 19, pp. 5386–5392, Oct. 1, 2020.

[116] P. Taheri, D. Menikkumbura, E. Vanini, S. Fahmy, P. Eugster, and T. Edsall, "RoCC," in *Proc. 16th Int. Conf. Emerg. Netw. Exp. Technol.*, Nov. 2020, pp. 17–30.

[117] C. Tian, B. Li, L. Qin, J. Zheng, J. Yang, W. Wang, G. Chen, and W. Dou, "P-PFC: Reducing tail latency with predictive PFC in lossless data center networks," *IEEE Trans. Parallel Distrib. Syst.*, vol. 31, no. 6, pp. 1447–1459, Jun. 2020.

[118] Y. Gilad, T. Hlavacek, A. Herzberg, M. Schapira, and H. Shulman, "Perfect is the enemy of good: Setting realistic goals for BGP security," in *Proc. 17th ACM Workshop Hot Topics Netw.*, Nov. 2018, pp. 57–63.

[119] W.-X. Liu, J. Cai, Y. Wang, Q. C. Chen, and J.-Q. Zeng, "Fine-grained flow classification using deep learning for software defined data center networks," *J. Netw. Comput. Appl.*, vol. 168, Oct. 2020, Art. no. 102766.

[120] A. Yu, H. Yang, K. K. Nguyen, J. Zhang, and M. Cheriet, "Burst traffic scheduling for hybrid E/O switching DCN: An error feedback spiking neural network approach," *IEEE Trans. Netw. Service Manage.*, vol. 18, no. 1, pp. 882–893, Mar. 2021.

[121] S. Luo, H. Yu, K. Li, and H. Xing, "Efficient file dissemination in data center networks with priority-based adaptive multicast," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 6, pp. 1161–1175, Jun. 2020.

[122] X. Shi, Y. Li, H. Xie, T. Yang, L. Zhang, P. Liu, H. Zhang, and Z. Liang, "An OpenFlow-based load balancing strategy in SDN," *Comput., Mater. Continua*, vol. 62, no. 1, pp. 385–398, 2020.

[123] Y. Liu, H. Gu, F. Yan, and N. Calabretta, "Highly-efficient switch migration for controller load balancing in elastic optical inter-datacenter networks," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 9, pp. 2748–2761, Sep. 2021.

[124] F. Wang, H. Yao, Q. Zhang, J. Wang, R. Gao, D. Guo, and M. Guizani, "Dynamic distributed multi-path aided load balancing for optical data center networks," *IEEE Trans. Netw. Service Manage.*, vol. 19, no. 2, pp. 991–1005, Jun. 2022.

[125] J. Huang, W. Lyu, W. Li, J. Wang, and T. He, "Mitigating packet reordering for random packet spraying in data center networks," *IEEE/ACM Trans. Netw.*, vol. 29, no. 3, pp. 1183–1196, Jun. 2021.

[126] Y. Lin, Y. Zhou, Z. Liu, K. Liu, Y. Wang, M. Xu, J. Bi, Y. Liu, and J. Wu, "NetView: Towards on-demand network-wide telemetry in the data center," *Comput. Netw.*, vol. 180, Oct. 2020, Art. no. 107386.

[127] M. S. Elsayed, N.-A. Le-Khac, and A. D. Jurcut, "InSDN: A novel SDN intrusion dataset," *IEEE Access*, vol. 8, pp. 165263–165284, 2020.

[128] M. Sutherland, S. Gupta, B. Falsafi, V. Marathe, D. Pnevmatikatos, and A. Daglis, "The NEBULA RPC-optimized architecture," in *Proc. ACM/IEEE 47th Annu. Int. Symp. Comput. Archit. (ISCA)*, May 2020, pp. 199–212.

[129] A. Hirsi, L. Audah, A. Salh, N. M. Sahar, S. Ahmed, and M. A. Alhartomi, "DDoS anomaly detection in software-defined networks: An evaluation of machine learning techniques for traffic classification and prediction," in *Proc. Int. Conf. Future Technol. Smart Soc. (ICFTSS)*, Aug. 2024, pp. 100–105, doi: 10.1109/icftss61109.2024.10691328.

[130] A. Hirsi, L. Audah, A. Salh, M. A. Alhartomi, and S. Ahmed, "Detecting DDoS threats using supervised machine learning for traffic classification in software defined networking," *IEEE Access*, vol. 12, pp. 166675–166702, 2024, doi: 10.1109/ACCESS.2024.3486034.

[131] A. H. Abdi, L. Audah, A. Salh, M. A. Alhartomi, H. Rasheed, S. Ahmed, and A. Tahir, "Security control and data planes of SDN: A comprehensive review of traditional, AI, and MTD approaches to security solutions," *IEEE Access*, vol. 12, pp. 69941–69980, 2024, doi: 10.1109/ACCESS.2024.3393548.

[132] A. Aral and I. Brandic, "Learning spatiotemporal failure dependencies for resilient edge computing services," *IEEE Trans. Parallel Distrib. Syst.*, vol. 32, no. 7, pp. 1578–1590, Jul. 2021.

[133] B. Cao, J. Zhao, P. Yang, Y. Gu, K. Muhammad, J. J. P. C. Rodrigues, and V. H. C. de Albuquerque, "Multiobjective 3-D topology optimization of next-generation wireless data center network," *IEEE Trans. Ind. Informat.*, vol. 16, no. 5, pp. 3597–3605, May 2020.

[134] K. Devi R., M. Gurusamy, and V. P., "An efficient cloud data center allocation to the source of requests," *J. Organizational End User Comput.*, vol. 32, no. 3, pp. 23–36, Jul. 2020.

[135] M. Lu, Y. Gu, and D. Xie, "A dynamic and collaborative multi-layer virtual network embedding algorithm in SDN based on reinforcement learning," *IEEE Trans. Netw. Service Manage.*, vol. 17, no. 4, pp. 2305–2317, Dec. 2020.
[136] Y. Liu, F. Zhou, C. Chen, Z. Zhu, T. Shang, and J.-M. Torres-Moreno, "Disaster protection in inter-DataCenter networks leveraging cooperative storage," *IEEE Trans. Netw. Service Manage.*, vol. 18, no. 3, pp. 2598–2611, Sep. 2021.
[137] M. Wang, B. Cheng, and J. Chen, "Joint availability guarantee and resource optimization of virtual network function placement in data center networks," *IEEE Trans. Netw. Service Manage.*, vol. 17, no. 2, pp. 821–834, Jun. 2020.
[138] S. A. Mamun, A. Ganguly, P. P. Markopoulos, A. Kwasinski, and M. Kwon, "Security vulnerabilities of server-centric wireless datacenters," in *Proc. IEEE Conf. Commun. Netw. Secur. (CNS)*, Jun. 2020, pp. 1–9.
[139] A. Ganji, A. Singh, and M. Shahzad, "Characterizing the impact of TCP coexistence in data center networks," in *Proc. IEEE 40th Int. Conf. Distrib. Comput. Syst. (ICDCS)*, Nov. 2020, pp. 388–398.
[140] W. Li, X. Yuan, W. Qu, H. Qi, X. Zhou, S. Chen, and R. Xu, "Efficient coflow transmission for distributed stream processing," in *Proc. IEEE Conf. Comput. Commun.*, Jul. 2020, pp. 1319–1328.
[141] M. Macktoobian, Z. Shu, and Q. Zhao, "Traffic divergence theory: An analysis formalism for dynamic networks," *IEEE Access*, vol. 12, pp. 67512–67524, 2024.
[142] R.-X. Hao, M.-M. Gu, and J.-M. Chang, "Relationship between extra edge connectivity and component edge connectivity for regular graphs," *Theor. Comput. Sci.*, vol. 833, pp. 41–55, Sep. 2020.
[143] L. Chen, Y. Feng, B. Li, and B. Li, "A case for pricing bandwidth: Sharing datacenter networks with cost dominant fairness," *IEEE Trans. Parallel Distrib. Syst.*, vol. 32, no. 5, pp. 1256–1269, May 2021.
[144] A. Ulhassan, T. Pham, Q. Wang, and J. Stewart, "Statistical method for multi-path interference detection in IMDD optical links," *J. Lightw. Technol.*, vol. 41, no. 14, pp. 4699–4704, Jul. 15, 2023.
[145] R. Matsumoto, R. Konoike, H. Matsuura, K. Suzuki, T. Inoue, Y. Mori, K. Ikeda, S. Namiki, and K.-I. Sato, "Design and verification of a LO bank enabled by fixed-wavelength lasers and fast tunable silicon ring filters for creating large scale optical switches," *Opt. Exp.*, vol. 29, no. 24, p. 39930, Nov. 2021.
[146] X. Lin, S. Yue, Y. Tan, W. Sun, M. Veeraraghavan, and W. Hu, "A partial Store-and-Forward scheduling method for inter-datacenter bulk data transfers," *IEEE Access*, vol. 8, pp. 128167–128181, 2020.
[147] C. Chen, J. Ye, Y. Gao, S. Liu, and Y. Xu, "HF^{2}T: host-based flowlet fine-tuning for RDMA load balancing," in *Proc. 8th Asia–Pacific Workshop Netw.*, Aug. 2024, pp. 9–15.
[148] A. Althoubi, R. Alshahrani, and H. Peyravi, "Delay analysis in IoT sensor networks," *Sensors*, vol. 21, no. 11, p. 3876, Jun. 2021.
[149] Y. Liang, M. Lu, Z.-J.-M. Shen, and R. Tang, "Data center network design for Internet-related services and cloud computing," *SSRN Electron. J.*, pp. 2077–2101, Jul. 2021.
[150] V. V. Spirin, J. L. Bueno Escobedo, D. A. Korobko, P. Mégret, and A. A. Fotiadi, "Dual-frequency laser comprising a single fiber ring cavity for self-injection locking of DFB laser diode and Brillouin lasing," *Opt. Exp.*, vol. 28, no. 25, p. 37322, Nov. 2020.
[151] S. Hu, W. Bai, G. Zeng, Z. Wang, B. Qiao, K. Chen, K. Tan, and Y. Wang, "Aeolus: A building block for proactive transport in datacenters," in *Proc. Annu. Conf. ACM Special Interest Group Data Commun. Appl., Technologies, Archit., Protocols Comput. Commun. (SIGCOMM '20)*, New York, NY, USA: ACM, Aug. 2020, p. 13, doi: 10.1145/3387514.3405878.
[152] C. H. Song, X. Z. Khooi, R. Joshi, I. Choi, J. Li, and M. C. Chan, "Network load balancing with in-network reordering support for RDMA," in *Proc. ACM SIGCOMM Conf.*, Sep. 2023, pp. 816–831.
[153] A. Pourhabibi, M. Sutherland, A. Daglis, and B. Falsafi, "Cerebros: Evading the RPC tax in datacenters," in *Proc. 54th Annu. IEEE/ACM Int. Symp. Microarchitecture*, Oct. 2021, pp. 407–420.
[154] G. Ranjan, T. N. Nguyen, H. Mekky, and Z.-L. Zhang, "On virtual id assignment in networks for high resilience routing: A theoretical framework," in *Proc. IEEE Global Commun. Conf.*, Dec. 2020, pp. 1–6.
[155] Z. Xu, C. Sun, T. Ji, J. H. Manton, and W. Shieh, "Feedforward and recurrent neural network-based transfer learning for nonlinear equalization in short-reach optical links," *J. Lightw. Technol.*, vol. 39, no. 2, pp. 475–480, Jan. 15, 2021.
[156] W.-X. Liu, J. Cai, Q. C. Chen, and Y. Wang, "DRL-R: Deep reinforcement learning approach for intelligent routing in software-defined data-center networks," *J. Netw. Comput. Appl.*, vol. 177, Mar. 2021, Art. no. 102865.
[157] P. Sun, Z. Guo, S. Liu, J. Lan, J. Wang, and Y. Hu, "SmartFCT: Improving power-efficiency for data center networks with deep reinforcement learning," *Comput. Netw.*, vol. 179, Oct. 2020, Art. no. 107255.
[158] K. A. Clark, D. Cletheroe, T. Gerard, I. Haller, K. Jozwik, K. Shi, B. Thomsen, H. Williams, G. Zervas, H. Ballani, P. Bayvel, P. Costa, and Z. Liu, "Synchronous subnanosecond clock and data recovery for optically switched data centres using clock phase caching," *Nature Electron.*, vol. 3, no. 7, pp. 426–433, Jun. 2020.
[159] L. Shalev, H. Ayoub, N. Bshara, and E. Sabbag, "A cloud-optimized transport protocol for elastic and scalable HPC," *IEEE Micro*, vol. 40, no. 6, pp. 67–73, Nov. 2020.
[160] F. Tian, Y. Zhang, W. Ye, C. Jin, Z. Wu, and Z.-L. Zhang, "Accelerating distributed deep learning using multi-path RDMA in data center networks," in *Proc. ACM SIGCOMM Symp. SDN Res. (SOSR)*, Oct. 2021, pp. 88–100.
[161] C. Jia, T. Pan, Z. Bian, X. Lin, E. Song, C. Xu, T. Huang, and Y. Liu, "Rapid detection and localization of gray failures in data centers via in-band network telemetry," in *Proc. IEEE/IFIP Netw. Operations Manage. Symp.*, Apr. 2020, pp. 1–9.
[162] C. K. Dominicini, G. L. Vassoler, R. Valentim, R. S. Villaca, M. R. N. Ribeiro, M. Martinello, and E. Zambon, "KeySFC: Traffic steering using strict source routing for dynamic and efficient network orchestration," *Comput. Netw.*, vol. 167, Feb. 2020, Art. no. 106975.
[163] H. Taka, T. Inoue, and E. Oki, "Design model of a twisted and folded clos network with multi-step grouped intermediate switches guaranteeing admissible blocking probability," *J. Opt. Commun. Netw.*, vol. 16, no. 3, p. 328, Feb. 2024.
[164] M. Wang, B. Cheng, and J. Chen, "Joint Availability- and traffic-aware placement of parallelized service chain in NFV-enabled data center," in *Proc. IEEE Int. Conf. Web Services (ICWS)*, Oct. 2020, pp. 216–223.
[165] M. Safdar, Y. Abbas, W. Iqbal, M. Y. Umair, and A. Wakeel, "ARP overhead reduction framework for software defined data centers," *J. Netw. Syst. Manage.*, vol. 30, no. 3, May 2022, Art. no. 50.
[166] G. Suila Kuaban, B. Singh Soodan, R. Kumar, and P. Czekalski, "Analysis of the performance of a cloud computing processing queue with correlated reneging of tasks and resubmission," in *Proc. Int. Conf. Electr., Comput. Energy Technol. (ICECET)*, Dec. 2021, pp. 1–8.
[167] A. Ranjith, P. Subhash, G. Nagaraju, M. Srinivas, and A. Prashanth, "Performance study of three different types of data center network architectures, NovaCube, BCube, and FatTree using NS-3 simulator," in *Proc. Int. Conf. Comput. Commun. Technol.* Cham, Switzerland: Springer, 2024, pp. 411–421.
[168] M. Murakami, H. Kubokawa, K. Sugiura, E. Oki, S. Okamoto, and N. Yamanaka, "Online parameter tuning of the flow classification method in the energy-efficient data center network 'HOLS,'" *J. Opt. Commun. Netw.*, vol. 12, no. 11, p. 344, Nov. 2020.
[169] X. Wu, Z. Wang, W. Wang, and T. S. E. Ng, "Augmented queue: A scalable in-network abstraction for data center network sharing," in *Proc. ACM SIGCOMM Conf.*, Sep. 2023, pp. 305–318.
[170] P. Dong, X. Lu, T. Huang, L. Chen, Y. Yang, and L. Zhang, "Predictive queue-based rate control for low latency in lossless data center networks," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 3, pp. 3428–3439, Jun. 2024.
[171] G. Drainakis, P. Baziana, and A. Bogris, "Scalable and low server-to-server latency data center network architecture based on optical packet inter-rack and intra-rack switching," *J. Opt. Commun. Netw.*, vol. 15, no. 11, p. 804, Nov. 2023.
[172] Y. Liu, W. Li, W. Qu, and H. Qi, "BULB: Lightweight and automated load balancing for fast datacenter networks," in *Proc. 51st Int. Conf. Parallel Process.*, Aug. 2022, pp. 1–11.
[173] X. Zuo, Q. Li, J. Xiao, D. Zhao, and Y. Jiang, "Drift-bottle," in *Proc. 18th Int. Conf. Emerg. Netw. Exp. Technol.*, Nov. 2022, pp. 337–348.
[174] X. Tao, X. Qian, L. Han, W. Fan, Y. Shi, X. Zhu, Z. Li, S. Wei, and R. Xu, "Key flow first prioritized flow scheduling strategy in multi-tenant data centers," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 3, pp. 3264–3277, Jun. 2024.
[175] K. Chitavisutthivong, S. Supittayapornpong, P. Namyar, M. Zhang, M. Yu, and R. Govindan, "Optimal oblivious routing with concave objectives for structured networks," *IEEE/ACM Trans. Netw.*, vol. 31, no. 6, pp. 2669–2681, Dec. 2023.

[176] B. Babayigit and B. Ulu, "Deep learning for load balancing of SDN-based data center networks," *Int. J. Commun. Syst.*, vol. 34, no. 7, Feb. 2021, Art. no. e4760.
[177] F. Lin, H. Wang, G. Chen, G. Zhou, T. Xu, D. Wei, L. Chen, Y. Lu, A. Qu, H. Shao, and H. Jiang, "Fast, scalable and robust centralized routing for data center networks," *IEEE/ACM Trans. Netw.*, vol. 31, no. 6, pp. 2624–2639, Dec. 2023.
[178] D. De Silva and D. Alahakoon, "An artificial intelligence life cycle: From conception to production," *Patterns*, vol. 3, no. 6, Jun. 2022, Art. no. 100489.
[179] R. Ben Said, Z. Sabir, and I. Askerzade, "CNN-BiLSTM: A hybrid deep learning approach for network intrusion detection system in software-defined networking with hybrid feature selection," *IEEE Access*, vol. 11, pp. 138732–138747, 2023.
[180] S. Satpathi, S. Deb, R. Srikant, and H. Yan, "Learning latent events from network message logs," *IEEE/ACM Trans. Netw.*, vol. 27, no. 4, pp. 1728–1741, Aug. 2019.
[181] A. Mozo, B. Ordozgoiti, and S. Gómez-Canaval, "Forecasting short-term data center network traffic load with convolutional neural networks," *PLoS ONE*, vol. 13, no. 2, Feb. 2018, Art. no. e0191939.
[182] J. Gao, N. Yaseen, R. MacDavid, F. V. Frujeri, V. Liu, R. Bianchini, R. Aditya, X. Wang, H. Lee, D. Maltz, M. Yu, and B. Arzani, "Scouts," in *Proc. Annu. Conf. ACM Special Interest Group Data Commun. Appl., Technol., Archit., Protocols Comput. Commun.*, Jul. 2020, pp. 1–21.
[183] S. Zhang, Y. Liu, W. Meng, Z. Luo, J. Bu, S. Yang, P. Liang, D. Pei, J. Xu, Y. Zhang, Y. Chen, H. Dong, X. Qu, and L. Song, "PreFix," in *Proc. ACM Int. Conf. Meas. Model. Comput. Syst.*, Jun. 2018, pp. 1–6.
[184] F. Yan, X. Xue, X. Guo, B. Pan, J. Wang, S. Zhang, E. Khani, G. Guelbenzu, and N. Calabretta, "Load balance algorithm for an OPSquare datacenter network under real application traffic," *J. Opt. Commun. Netw.*, vol. 12, no. 8, p. 239, Aug. 2020.
[185] B. Nougnanke, Y. Labit, M. Bruyere, U. Aïvodji, and S. Ferlin, "ML-based performance modeling in SDN-enabled data center networks," *IEEE Trans. Netw. Service Manage.*, vol. 20, no. 1, pp. 815–829, Mar. 2023.
[186] C. Guo, L. Yuan, D. Xiang, Y. Dang, R. Huang, D. Maltz, Z. Liu, V. Wang, B. Pang, H. Chen, Z.-W. Lin, and V. Kurien, "Pingmesh," in *Proc. ACM Conf. Special Interest Group Data Commun.*, Aug. 2015, pp. 139–152.
[187] Y. Han, J.-H. Yoo, and J. W.-K. Hong, "Poisson shot-noise process based flow-level traffic matrix generation for data center networks," in *Proc. IFIP/IEEE Int. Symp. Integr. Netw. Manage. (IM)*, May 2015, pp. 450–457.
[188] N. Lin, Q. Zhao, L. Zhao, A. Hawbani, L. Liu, and G. Min, "A novel cost-effective controller placement scheme for software-defined vehicular networks," *IEEE Internet Things J.*, vol. 8, no. 18, pp. 14080–14093, Sep. 2021.
[189] W. Li, Y. Jiang, X. Zhang, F. Dang, F. Gao, H. Wang, and Q. Fan, "Reliability assurance dynamic SSC placement using reinforcement learning," *Information*, vol. 13, no. 2, p. 53, Jan. 2022.
[190] S. Li, Y. Qin, Z. Jiang, and W. Yang, "Efficient communication scheduling for parameter synchronization of DML in data center networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 4, pp. 1970–1985, Jul. 2022.
[191] Z. Guo, J. Wang, S. Liu, J. Ren, Y. Xu, and C. Yao, "Congestion-aware critical gradient scheduling for distributed machine learning in data center networks," *IEEE Trans. Cloud Comput.*, vol. 11, no. 3, pp. 2296–2311, Jul. 2023.
[192] P. Nallusamy, T. R. Reshmi, and M. Krishnan, "AGFT: Adaptive entries aggregation scheme to prevent overflow in multiple flow table environment," *Concurrency Comput., Pract. Exper.*, vol. 34, no. 1, Jan. 2022, Art. no. e6491.
[193] Y. Yang, Y. Wu, Y. Jiang, and Y. Shi, "One-bit Byzantine-tolerant distributed learning via Over-the-Air computation," *IEEE Trans. Wireless Commun.*, vol. 23, no. 6, pp. 5441–5455, Jun. 2024.
[194] X. Zhao, C. Wu, and X. Zhu, "Dynamic flow scheduling for DNN training workloads in data centers," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 6, pp. 6643–6657, Dec. 2024.
[195] T. Wang, X. Fan, K. Cheng, X. Du, H. Cai, and Y. Wang, "Parameterized deep reinforcement learning with hybrid action space for energy efficient data center networks," *Comput. Netw.*, vol. 235, Nov. 2023, Art. no. 109989.
[196] Q. Fu, E. Sun, K. Meng, M. Li, and Y. Zhang, "Deep Q-learning for routing schemes in SDN-based data center networks," *IEEE Access*, vol. 8, pp. 103491–103499, 2020.
[197] J. Hu, Z. Zhou, and J. Zhang, "Lightweight automatic ECN tuning based on deep reinforcement learning with ultra-low overhead in datacenter networks," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 6, pp. 6398–6408, Dec. 2024.
[198] S. Abdollahi, H. Asadi, A. Montazerolghaem, and S. M. Mazinani, "FMap: A fuzzy map for scheduling elephant flows throughjumping traveling salesman problemvariant towardsoftware-defined networking-baseddata center networks," *Concurrency Comput., Pract. Exper.*, vol. 35, no. 26, Jun. 2023, Art. no. e7841.
[199] X. Dong, L. Nie, Z. Liu, and Y. Xiang, "Slark: A performance robust decentralized inter-datacenter deadline-aware coflows scheduling framework with local information," *IEEE Trans. Parallel Distrib. Syst.*, vol. 36, no. 2, pp. 1–15, Feb. 2025.
[200] S. Abdollahi, A. Deldari, H. Asadi, A. Montazerolghaem, and S. M. Mazinani, "Flow-aware forwarding in SDN datacenters using a Knapsack-PSO-based solution," *IEEE Trans. Netw. Service Manage.*, vol. 18, no. 3, pp. 2902–2914, Sep. 2021.
[201] Y. Liu, J. Zhang, W. Li, Q. Wu, and P. Li, "Load balancing oriented predictive routing algorithm for data center networks," *Future Internet*, vol. 13, no. 2, p. 54, Feb. 2021.
[202] M. J. Abdel-Rahman, E. A. Mazied, F. Hassan, K. Teague, A. B. Mackenzie, S. F. Midkiff, K. V. Cardoso, and D. S. Nikolopoulos, "On robust optimal joint deployment and assignment of RAN intelligent controllers in O-RANs," *IEEE Open J. Commun. Soc.*, vol. 5, pp. 2358–2376, 2024.
[203] A. Arnaz, J. Lipman, M. Abolhasan, and M. Hiltunen, "Toward integrating intelligence and programmability in open radio access networks: A comprehensive survey," *IEEE Access*, vol. 10, pp. 67747–67770, 2022.
[204] C. Jiang, Y. Qiu, H. Gao, T. Fan, K. Li, and J. Wan, "An edge computing platform for intelligent operational monitoring in Internet data centers," *IEEE Access*, vol. 7, pp. 133375–133387, 2019, doi: 10.1109/ACCESS.2019.2939614.
[205] S. Chen, G. Zhang, S. S. Yu, Y. Mei, and Y. Zhang, "A review of isolated bidirectional DC–DC converters for data centers," *Chin. J. Electr. Eng.*, vol. 9, no. 4, pp. 1–22, Dec. 2023, doi: 10.23919/cjee.2023.000034.
[206] M. M. Than, "Resource management for minimizing energy and cost of geo-distributed data centers," *Walailak J. Sci. Technol. (WJST)*, vol. 18, no. 13, Jun. 2021, Art. no. 9619, doi: 10.48048/wjst.2021.9619.
[207] A. Poobalan, S. Sangeetha, and P. Shanthakumar, "Optimal switching and load distribution strategy for energy minimization and performance optimization of cloud data center," *J. Eng. Res.*, vol. 43, Nov. 2021, Art. no. 101013, doi: 10.36909/jer.13245.
[208] A. Kellerman, "Autonomous technologies for daily personal mobilities," *Eur. J. Geography*, vol. 14, no. 3, pp. 89–96, Aug. 2023, doi: 10.48088/ejg.a.kel.14.3.089.096.
[209] M. Aghaei, M. Kolahi, A. Nedaei, N. S. Venkatesh, S. M. Esmailifar, A. M. M. Sizkouhi, A. Aghamohammadi, A. K. V. Oliveira, A. Eskandari, P. Parvin, J. Milimonfared, V. Sugumaran, and R. Rüther, "Autonomous intelligent monitoring of photovoltaic systems: An In-depth multidisciplinary review," *Prog. Photovoltaics: Res. Appl.*, vol. 33, no. 3, pp. 381–409, Nov. 2024, doi: 10.1002/pip.3859.
[210] M. S. Hoosain, B. S. Paul, S. Kass, and S. Ramakrishna, "Tools towards the sustainability and circularity of data centers," *Circular Economy Sustainability*, vol. 3, no. 1, pp. 173–197, Mar. 2023.
[211] S. Mondal, F. B. Faruk, D. Rajbongshi, M. M. K. Efaz, and M. M. Islam, "GEECO: Green data centers for energy optimization and carbon footprint reduction," *Sustainability*, vol. 15, no. 21, p. 15249, Oct. 2023.
[212] S. Sarkar, A. Naug, R. Luna, A. Guillén, V. Gundecha, S. Ghorbanpour, S. Mousavi, D. Markovikj, and A. R. Babu, "Carbon footprint reduction for sustainable data centers in real-time," in *Proc. Conf. AAAI Artif. Intell.*, Mar. 2024, vol. 38, no. 20, pp. 22322–22330.
[213] Senthilkumar, Rajendran, Suresh, N. H. A. Rufus, R. C. Tanguturi, and R. S. Solanki, Tanguturi, and R. S. Solanki, "Computational engineering based approach on artificial intelligence and machine learning-driven robust data centre for safe management," *J. Mach. Comput.*, pp. 465–474, Oct. 2023.
[214] M. Manganelli, A. Soldati, L. Martirano, and S. Ramakrishna, "Strategies for improving the sustainability of data centers via energy mix, energy conservation, and circular energy," *Sustainability*, vol. 13, no. 11, p. 6114, May 2021.

[215] H. H. Mahmoud, M. Kadhim, M. Al-Shammari, I. M. Hameed, I. A. Barazanchi, R. Sekhar, P. Shah, and N. Solke, "Eco-friendly and secure data center to detection compromised devices utilizing swarm approach," *Int. J. Innov. Eng. Sci. (IJIES)*, vol. 17, no. 3, pp. 102–115, Jun. 2024.
[216] C. Seifert, "The barriers for voluntary environmental management systems—The case of EMAS in hospitals," *Sustainability*, vol. 10, no. 5, p. 1420, May 2018, doi: 10.3390/su10051420.

KOGUL SRIKANDABALA (Graduate Student Member, IEEE) received the B.Sc. degree in computer science and engineering from the University of Moratuwa, Sri Lanka. He is currently pursuing the Ph.D. degree with the La Trobe Business School, La Trobe University, Melbourne, Australia, undertaking an Industry Ph.D. in partnership with NEXTDC Ltd. He is also a Researcher with the Center for Data Analytics and Cognition (CDAC). Prior to this, he was a Lecturer (on contract) with the Department of Computer Science and Engineering, University of Moratuwa. His research interests include network optimization and the application of artificial intelligence in healthcare. His work focused on integrating data-driven methodologies with computational intelligence to address complex optimization challenges in healthcare and enterprise network systems.

KIRISHNNI PRABAGAR (Graduate Student Member, IEEE) received the B.Sc. degree (Hons.) in computer science and engineering from the University of Moratuwa, Sri Lanka. She is currently pursuing the Ph.D. degree with La Trobe University, undertaking an Industry Ph.D. in partnership with NEXTDC Ltd. Her research is centered on leveraging artificial intelligence, machine learning, and data science to drive intelligent optimization and adaptive resource management within network infrastructure. Before commencing her Ph.D. studies, she was a Lecturer with the Department of Computer Science and Engineering, University of Moratuwa. Her research interests include AI-driven optimization techniques for large-scale computing environments, focusing on predictive analytics, resource management, and intelligent automation within network infrastructure ecosystems.

SHALINKA JAYATILLEKE is currently a Lecturer in business analytics and artificial intelligence (AI) and the Course Coordinator of the Bachelor of Business Analytics Degree Program, La Trobe University, and a Researcher specializing in AI, data analytics, computer networks, and personalization. Her work has been published in leading peer-reviewed journals and conferences, and she has collaborated with industry and academia to drive innovation in AI and data-driven decision-making. Her research interests include AI-driven anomaly detection, hyper-personalization, and conversational AI, with applications in smart grids, digital consumer experience, and business intelligence.

PEARLIE ZHANG received the B.E. degree in computer engineering and commerce from the University of New South Wales, in 2019. In 2017, she joined NEXTDC Ltd., where she is currently a Network Operations Team Leader. Her research interests include network automation and system integration.

RICHARD ELLERBROCK received the degree in electrical engineering from Tshwane University of Technology. He is currently a Network Development Engineer with the Network Operations Team, NEXTDC Ltd. He has more than 30 years of global experience in data networks, communications, security, and software and platform development in various industries, including utilities, transport, managed services, financial services (equity exchanges), and data centers.

SEAN RINAS received the Bachelor of Electrical Engineering degree from the University of Saskatchewan, in 1997. He is currently the Head of Network Operations of NEXTDC Ltd. With a career spanning, since 1995, he has extensive experience in telecommunications architecture, delivery, and operations across Canada, USA, and Australia.

DASWIN DE SILVA (Senior Member, IEEE) received the Ph.D. degree in AI from Monash University, in 2011, focused on the algorithmic development of incremental learning without forgetting and bias. He is currently a Professor of artificial intelligence and analytics with La Trobe University, Australia. His expertise is in AI, algorithms, systems, applications, and ethics. He has co-authored more than 150 research articles, received AU$ more than 14 million in industry and government funding for AI research, and supervised 15 Ph.D.'s until completion. In the IEEE, he is the Chair of the Technical Committee on Responsible AI, a Governing Board Member of the IEEE Blockchain, and the IEEE AI Coalition. He is an editor of four journals and has been consulted by the ABC News, Channel 7 News, The Australian, Forbes, Cosmos, Times Higher Education, and The Conversation for his AI expertise.

DAMMINDA ALAHAKOON (Member, IEEE) is currently a Professor of business analytics and artificial intelligence and the Director of the Research Centre for Data Analytics and Cognition, La Trobe University. His research has been adopted in a range of domains from healthcare, utilities, smart cities, education, industrial optimization, and national security. His work has been published in more than 200 peer-reviewed articles and has generated more than AU$ 6 million in research and industry grants. His research is focused on the areas of machine learning, artificial intelligence, advanced data analytics, business intelligence, and the harnessing of such theories for practical tools and innovative technology for industry.